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When corporate risk management amplifies risks: Evidence from Europe's energy crisis in 2022

Alessandro Giannozzi [a], Antonia Kirilova [b,*]

[a] University of Florence, Via delle Pandette 32, 50127 Florence, Italy
[b] CUNEF Universidad, Calle Almansa 101, 28040 Madrid, Spain

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ABSTRACT

Derivatives are important risk management tools for hedging price risks but can expose firms to the risk of margin calls in times of market stress. This study investigates the impact of large margin calls on companies' financial outcomes. We document that most companies meet surging collateral requirements through borrowing, but firms without preexisting relationships with lenders resort to asset sales. Importantly, we show that margin calls have significant negative consequences for companies' default risk, cost of debt, and profit margins, and some of these effects persist for more than a year after the margin calls. Our evidence derives from the European energy crisis in 2022, when energy corporations had to deposit significant amounts of cash to cover margin calls related to commodity hedges. However, the results may be generalized to other markets and highlight the need for corporate risk management to consider margin and liquidity risks tied to derivatives use.

1. Introduction

The intrinsic presence of uncertainty in economic activities underscores the importance of risk management. For many corporations that navigate complex and unpredictable markets, the hedging of financial risks forms an integral part of their operations. While hedging practices undoubtedly bring benefits in terms of corporate resilience as well as overall market stability, in certain circumstances they may inadvertently amplify risks. This study aims to explore some of the unintended negative consequences that corporate hedgers may face in times of market stress. Specifically, we show that margin calls triggered by fluctuations in the value of derivatives positions negatively impact companies' default risk, cost of debt, and profit margins, and these effects can persist even after markets stabilize.

In a theoretical world with perfect capital markets, such as in Modigliani and Miller (1958), risk management does not matter. However, in reality, imperfect capital markets make it essential for companies to hedge risks. Theoretical research proposes that risk management can have various value-adding benefits and, given different conditions at different firms, predicts heterogeneity in their hedging strategies (see e.g., Smith and Stulz, 1985; Froot et al., 1993). Empirical evidence by Bartram, Brown and Conrad (2011), Chen and King (2014), Gilje and Taillard (2017), and others, supports the notion that hedging with derivatives offers numerous benefits for firms: it increases firm value, reduces bankruptcy risk and overall firm risk, reduces the cost of debt, and mitigates underinvestment.

Despite the various benefits of hedging, there is one key drawback of using derivatives. Exchange-traded derivatives (ETD) contracts are marked to market, with profits and losses settled on a daily basis, and exchanges require both counterparties to deposit liquid funds in a margin account to cover potential future losses on their positions. Counterparties post initial margins to open positions, and in times of market volatility may need to deposit significant amounts of cash to cover variation margins. Over the past decade, over-the-counter (OTC)

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* Corresponding author.
E-mail addresses: alessandro.giannozzi@unifi.it (A. Giannozzi), antonia.kirilova@cunef.edu (A. Kirilova).

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derivatives have increasingly incorporated bilateral exchanges of liquid collateral as well. These collateral requirements tie up a firm's liquid resources, rendering them unavailable for other uses. Firms' ex ante decision about how much to hedge based on their collateral constraints is examined by several studies (see e.g., Rampini and Viswanathan, 2010; Rampini, Sufi, and Viswanathan, 2014). However, to our knowledge, there is no prior research examining the effects of sudden increases in collateral requirements due to losses on existing derivative hedges.

Spikes in collateral or margin requirements,[1] and the resulting margin calls, tend to be more frequent and severe in times of market stress and heightened price volatility. Typically, the cash has to be deposited within a short deadline of one to five days from the margin call. If the firm fails to do so, the exchange (or the non-defaulting counterparty in bilateral OTC contracts) has the right to close outstanding positions, forcing the loss to be realized before maturity, thus exposing the firm to potential distress or in extreme cases even bankruptcy. For this reason, the firm would typically try to maintain their derivatives positions by making timely collateral payments. If market prices eventually return to normal levels, the cash is returned to the company. Therefore, in theory, margin calls should not have a lasting impact on corporations. However, in practice, we show that this is not necessarily the case.

When companies face substantial margin calls triggered by mark-to-market losses on volatile derivatives positions, they are forced to either deplete their internal cash reserves, sell assets, or seek credit from external sources to meet the collateral requirements. As a result, the liquidity stress may lead to financial instability. Moreover, it may raise concerns among investors and lenders about the company's risk management practices and overall financial health. Therefore, we hypothesize that large margin calls may have lasting detrimental effects on companies' default probabilities, borrowing costs, and profitability.

To test this hypothesis, we use Europe's energy crisis in 2022 as the backdrop for our study. There are three main reasons why it provides an excellent setting for our empirical analyses. The first reason relates to the choice of an industry where derivatives are widely used for hedging and have a high potential impact on companies' financial health. In general, there are three broad categories of risk which corporations can hedge against through the use of derivative instruments: interest rate risk, foreign exchange risk, and commodity price risk. While the hedging of interest rate and foreign exchange risks can be important, it is unlikely to produce economically significant effects for most company portfolios (Guay and Kothari, 2003). On the other hand, commodity producers face significant exposure to commodity price risks, which makes risk management imperative and hedging a central part of their operations. The prevalence of hedging among commodity producers is reflected in the unprecedented growth of commodity derivatives markets over the past decade. According to data by the World Federation of Exchanges (WFE), the volume of exchange-traded commodity derivatives rose from almost 4 billion contracts traded in 2014 to more than 11 billion in 2023.[2] In the European Union (EU) markets, the European Securities and Markets Authority (ESMA) reveals that notional amounts of commodity derivatives outstanding at the end of 2021 reached more than EUR 2.5 trillion, and grew further to EUR 3.3 trillion by the end of 2022.[3] Almost 40 % of these amounts are traded by non-financial firms, such as the energy companies in our sample. Energy firms face substantial risk regarding the prices they receive for their production of oil, gas, and electricity. Most energy producers and distributors use derivative instruments to hedge against unfavorable swings in oil, gas, and/or power prices.[4] Typically they pre-sell future output one or more years ahead of time using short positions in derivatives. This secures a minimum price for their output, protecting cash flows in times of declining energy prices. However, in times of rising prices, energy firms tend to be the losing counterparties in derivatives contracts and may suffer margin calls.

The second reason to examine hedging in energy commodities is that margin requirements for these derivatives are much higher compared to others due to the high volatility in energy markets. In particular, the historical annualized volatility from 2015 to 2023 of Brent crude oil futures is 41 %, that of TTF natural gas futures is 75 %, and that of Dutch power futures is 62 %. As a comparison, the most traded currency pair in the world, the EUR/USD, has an annualized volatility of only 8 % during the same historical period.[5] European energy markets began exhibiting volatility in the second half of 2021, although at that time it was still manageable for firms. However, the sharp increases in energy prices following Russia's invasion of Ukraine on 24 February 2022 led to huge margin calls for many European energy firms, leaving them unable to find cash fast enough and badly exposed to bankruptcy risk. Chart A.3 in the Financial Stability Review published by the European Central Bank (ECB) in November 2022[6] illustrates the increases in margin requirements that accompanied the price surges. Multiple companies reveal in their financial statements or in earnings calls that their margin deposits reached billions of euros in 2022. It was estimated that the industry of European power firms needed more than 1.5 trillion euros to cover the collective cost of margin calls, and that estimate did not include British firms.[7] These large margin calls created sudden liquidity shortages and threatened the stability of numerous institutions.

The third reason for focusing our study on Europe's 2022 energy crisis is that it highlighted the serious consequences of risk amplification and drew the attention of policymakers. Following the 2008 Global Financial Crisis, authorities worldwide have tightened the regulation of derivatives markets, imposing central clearing and stricter margin requirements. While these generally help to minimize counterparty default risk, they can generate significant liquidity risk in times of market stress. The recent energy crisis and accompanying margin calls have led to regulatory discussions about central clearing and margining practices in derivatives markets, and potential ways to prevent such issues from arising again and threatening financial stability. With our study, we aim to contribute to the debate by highlighting how companies may be affected by these policies.

We start by constructing a novel proxy for the size of margin calls experienced by each firm based on balance sheet data. Then, we proceed with an investigation of how firms facing margin calls deal with the liquidity demands. We show that margin deposits remain elevated throughout the period of margin calls, indicating that most firms maintained their derivatives positions. We show that, in general, credit in various forms, including short-term and long-term bank debt, term loans, and revolving credit lines, seems to be the main source of liquid funds for most firms exposed to margin calls during the crisis. However, borrowing does not appear to be a viable option for all. Smaller firms which may have weaker relationships with lenders, as well as firms without availability of pre-approved revolving credit, resort to asset sales, as evidenced by a drop in their fixed assets. Internal cash reserves

[1] We use the terms collateral and margins interchangeably when referring to the context of derivatives trading.

[2] https://focus.world-exchanges.org/statistics

[3] https://www.esma.europa.eu/document/eu-derivatives-markets-2023

[4] Although derivatives can also be used for speculation (through naked positions) rather than hedging, Bartram (2019) finds that this is relatively rare among non-financial firms.

[5] The reported volatilities are based on own calculations using daily data from S&P Capital IQ and Bloomberg.

[6] https://www.ecb.europa.eu/pub/financial-stability/fsr/special/html/ecb.fsrart202211_01~173476301a.en.html

[7] https://www.reuters.com/business/energy/wide-demand-reduction-only-feasible-solution-europe-energy-crisis-equinor-2022-09-06/

do not seem to be a major source of liquidity for stressed firms, except for large companies which are most likely to have existing cash buffers. Regardless of how firms cover margin calls, it has a clear impact on their balance sheets, with many firms becoming more indebted and some reducing their asset base.

Next, we examine the short-term and long-term impacts of margin calls on companies' financial health, stability, and performance. We implement Difference-in-Differences (DiD) analyses to study changes in firm outcomes after the energy market turmoil, depending on the size of margin calls they suffer. Our tests reveal significant effects of margin calls on several firm-level outcome variables.

First, we find strong evidence that experiencing a margin call increases a firm's credit risk and probability of default, as reflected by elevated Credit Default Swaps (CDS) spreads after the market crisis for firms with a larger Margin Calls Proxy. The effect is strongest in the months immediately after the shock, but it remains significant for at least a year afterwards. In terms of economic significance, a one-standard-deviation increase in the size of margin calls experienced results in a 10.6 %−15.7 % higher CDS spread in the following 10 days and 1.3 %−6.5 % higher spread in the following year.

Second, we examine how the market's perception of elevated default risk spills over to creditors' estimations of firms' creditworthiness as reflected by their borrowing costs. While the effect of margin calls on the cost of debt is not significant for large firms and firms with large credit lines available, we do uncover significant increases in the cost of debt for affected small firms and firms with restricted access to credit lines. Even though the debt levels of all companies rise as a result of experiencing margin calls, those who have existing relationships with lenders obtain the necessary liquid funds more quickly and at pre-negotiated rates, thus reducing the overall cost and negative impacts of the liquidity shock. On the other hand, firms which are less likely to have preexisting relationships with lenders or pre-approved credit lines are more susceptible to the long-term negative effects of margin calls. Those of them that experience larger margin calls face more severe increases in their borrowing costs in subsequent quarters, and this effect persists for more than a year after the crisis. These results are highly economically significant: In the subsample of small firms (firms with small credit lines available), a one-standard-deviation increase in margin calls size results in 43 % (50 %) higher cost of debt subsequently. This impressive rise in borrowing costs reflects creditors' perception of these firms as more financially vulnerable following the liquidity shocks, rendering them as higher-risk borrowers with an uncertain ability to meet obligations if faced with further market volatility.

Finally, we show that there is a short-term drop in net profit margins for companies that experience large margin calls. If hedgers are aware of their exposure to margin call risk and factor it into their risk management strategies, they will be sufficiently prepared, and margin calls should not impact their competitive positions or profitability. Yet, we find that this is not the case in our sample. As a consequence of the margin calls, large firms in particular exhibit a decline in profit margins. Although the drop in profitability appears only in the quarter immediately after the most severe margin calls, it indicates that firms did not fully anticipate the liquidity shocks that they experienced in the 2022 energy crisis. This event may serve as a cautionary tale for firms to consider the risk of margin calls when devising their hedging strategies.

Overall, we aim to highlight that although the use of derivatives to hedge price risks generally reduces the probability of financial distress, it can produce the opposite effect in times of market stress. The results of our analysis reveal significant negative impacts of margin calls on companies' default risk, cost of debt, and bottom lines, with some of these consequences lasting for more than a year after market prices subside. The reported effects are not driven by common trends or fixed effects. While the European energy crisis serves as a useful empirical setting for our study, the results we draw from it may be generalizable to any market where companies are exposed to the risk of margin calls.

To the best of our knowledge, this is the first study to document the negative impacts of margin calls on companies' financial health and performance. We contribute to existing literature in several ways. First, we add to the literature related to the impact of hedging on corporations by analyzing the potential negative side effects of hedging. Second, we contribute to the default risk literature by highlighting a unique source of risk, namely margin call risk, which can threaten firm stability when markets become stressed. We suggest the possibility to test and incorporate this risk factor into default risk models (see e.g., Merton, 1974; Jarrow and Turnbull, 1995; Altman, 1968, 2013, 2018; and others). Third, our study implies that liquidity risk may not be fully captured by traditional measures such as liquidity ratios but may also be related to the risk of margin calls.

Finally, our results have important managerial and regulatory implications. They emphasize the need for corporate hedgers to continuously evaluate exposure to margin risk and maintain liquidity preparedness. From a regulatory perspective, streamlining margining practices and improving transparency of margin models could allow derivatives market participants to predict margin requirements more accurately and anticipate their liquidity needs. With rising geopolitical tensions and persistent market uncertainty in recent years, crises could resurface, stressing the importance of continued careful monitoring and assessment of risks. Effective coordination between corporate treasurers and policymakers is thus essential to ensure that risk management practices continue to enhance, rather than undermine, financial resilience.

The rest of this study is organized as follows. Section 2 describes the market setting that we use for our study and provides details about the data and methodology. Section 3 reports the empirical findings of our analyses. Section 4 discusses the implications of our results. Section 5 concludes.

2. Background, data and methodology

2.1. Europe's energy crisis in 2022

European energy markets experienced unprecedented turbulence during 2022. To identify the periods with the largest price jumps, we plot daily energy prices from July 2020 until December 2023 in Fig. 1. Data about oil prices is from S&P Capital IQ and data about gas and power prices is from Bloomberg. We choose the main commodity contracts which typically serve as benchmarks for European energy prices. In particular, oil prices refer to Brent Crude Oil futures contracts traded on the Intercontinental Exchange (ICE), natural gas prices refer to ICE TTF futures contracts, and electricity prices refer to Dutch Power Base Load Futures on the ICE.

Gas and power prices started to increase towards the end of 2021, while oil prices remained relatively stable at that time. After the Russian invasion of Ukraine on 24 February 2022, all European energy markets experienced sudden extraordinary spikes in prices and volatility. The largest daily returns in our sample were seen during the first quarter of 2022: 51.1 %, 46.2 %, and 8.3 % in the gas, electricity, and oil markets, respectively. Energy companies with existing positions in energy derivatives would have faced sudden re-valuations of their contract holdings. Since many energy producers pre-sell their future output by entering the short side of derivatives contracts, they would be the losing counterparty in periods of price increases. Therefore, we can expect them to have experienced sudden severe margin calls in quarter one of 2022. Corroborating this timeline, Chart A.3 in the ECB Financial Stability Review shows large spikes in both initial and variation margins posted by euro area entities on portfolios containing commodity derivatives precisely around that time. Although markets subsided during the following few months, they were headed for a second wave of turbulence in quarter three of 2022. While oil prices never returned to their March 2022 maximum of 117.5 euros per barrel, gas and power prices reached new peaks on 26 August 2022, at 339.2 euros and 644.2 euros per megawatt-hour, respectively. In a risk analysis report ESMA writes:

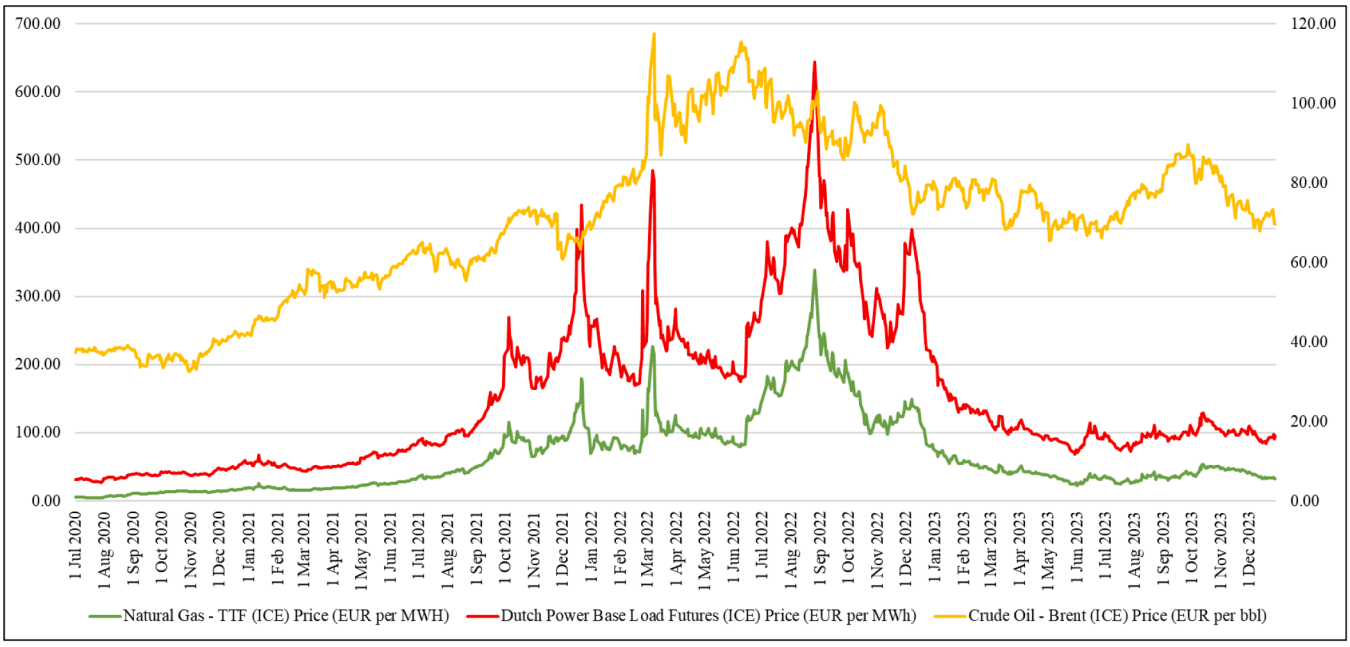


Fig. 1. Commodity Prices: July 2020 - December 2023. This figure shows the evolution of commodity prices in the period from July 2020 to December 2023 for the main commodity futures contracts in the European energy markets. In particular, prices of natural gas refer to TTF contracts on the Intercontinental Exchange (ICE); electricity prices are for Dutch Power Base Load Futures on the ICE; finally, prices of crude oil refer to Brent oil contracts on the ICE. Gas and electricity prices are plotted on the primary axis, and oil prices are plotted on the secondary axis.

"On this day, the largest 1-day margin increase was recorded, 11 % for all commodities CCPs."[8] Once again, the ECB report provides evidence that initial and variation margins rose sharply together with market prices and reached extreme levels in the third quarter of 2022.

A company facing margin calls needs to either come up with the necessary cash to top up their margin account,[9] or close its derivatives positions thus realizing losses that may reach devastating amounts. The reports by the ECB and ESMA mention that, despite the difficult market environment in 2022, most ETD positions were maintained, with only a slight shift to OTC trading towards the end of the year, and margin calls were met on time. In theory, cash collateral is deposited temporarily, and as long as market prices return to normal levels, it is returned to the depositor. However, to be able to cover the massive liquidity demands in 2022, European energy companies would have had to borrow large amounts of debt, sell assets, and/or use up their internal cash reserves thus potentially missing out on investment opportunities. Despite the fact that energy prices started to subside in the last quarter of 2022 and continued to decline in 2023, meaning that margin deposits were eventually returned to firms, the means used to obtain the cash in the first place may have had lasting destabilizing effects on their business operations and financial health. This is the main hypothesis of our study and in Section 3 we provide evidence consistent with it.

2.2. Data and sample selection

We use firm-level data provided by S&P Capital IQ. Our initial sample includes all European publicly listed companies with primary operations in the energy and utilities industries (except water utilities) as classified by S&P Capital IQ. We further refine the sample by selecting companies classified into any of the following industries: 'Crude Petroleum and Natural Gas', 'Drilling Oil and Gas Wells', 'Oil and Gas Field Exploration Services', 'Oil and Gas Field Services, NEC', 'Petroleum Refining', 'Miscellaneous Products Of Petroleum and Coal', 'Electric, Gas and Sanitary Services', 'Electric Services', 'Natural Gas Transmission', 'Natural Gas Transmission and Distribution', 'Natural Gas Distribution', 'Electric and Other Services Combined', 'Gas and Other Services Combined', 'Cogeneration Services and Small Power Producers'.[10] We require the country of incorporation to be in Europe to ensure that all firms in the sample follow the International Financial Reporting Standards (IFRS). To exclude missing or irrelevant observations, we implement several additional filters which are detailed in Appendix A1. Bankrupt firms are excluded from our analyses. Finally, we exclude the German energy company Uniper from our sample because it received a government bailout and was nationalized in the second half of 2022, and this direct cash injection by the government would mask the true impact of margin calls on Uniper's financial results.[11] The final sample contains 120 unique companies.

The sample period for our tests covers three time periods: i) a period of generally stable energy prices before the margin calls, covering three quarters from 2020-Q4[12] until 2021-Q2; ii) the period of market turbulence and margin calls, starting from 2021-Q3 and extending until 2022-Q3; and iii) the period after the most severe margin calls when prices start to subside, extending from the end of 2022-Q3 until the end of 2023-Q4.[13]

[8] https://www.esma.europa.eu/sites/default/files/2023-10/ESMA50-524821-2963_TRV_Article_the_August_2022_surge_in_the_price_of_natural_gas_futures.pdf

[9] According to the ECB, while initial margins could be posted in either cash or collateral in the form of high-quality securities, variation margins could be paid only in cash.

[10] SIC Codes: 1311, 1381, 1382, 1389, 2911, 2990, 4900, 4911, 4922, 4923, 4924, 4931, 4932, 4991.

[11] The results remain robust if we include Uniper in the sample. For details about Uniper's government aid, see: https://www.bloomberg.com/news/articles/2022-07-22/germany-bails-out-uniper-in-fallout-from-russian-gas-squeeze ; and https://www.reuters.com/business/energy/germany-completes-nationalization-uniper-ministries-2022-12-22/

[12] We do not extend the sample further back in order to avoid capturing any effects of the COVID-19 pandemic-related market shock.

[13] Most firms have a fiscal year-end at the end of December. For firms that do not, we align their fiscal periods to the calendar quarters for consistent analysis.

2.3. Estimating margin calls

We begin by constructing a novel proxy for margin calls size from balance sheet data. While some firms may report the exact amounts of collateral requirements they are subject to, this information is not mandatory to report and details about margin deposits tend to be scarce in financial statements. Nevertheless, we identify three variables reported in S&P Capital IQ's standardized Balance Sheets whose fluctuations help us to plausibly estimate the size of margin calls experienced by each firm. As discussed earlier, cash deposited as collateral for potential future losses on derivatives positions is eventually returned to its owner if losses are not realized. Therefore, it is a receivable for the depositing firm. Since margin deposits do not fall within the traditional definition of Accounts Receivable or Notes Receivable, they are commonly listed as Other Receivables. This is the first variable we use in the calculation of our proxy. According to S&P Capital IQ's definition of Other Receivables, they include margin and capital markets receivables, derivative related receivables, and securities clearing accounts receivable. The second variable we use is Restricted Cash, which reflects any cash that is reserved or restricted for use or transfer, such as when an exchange requires a company to set aside a specific amount as collateral, thereby making it unavailable for other uses. Finally, the third variable is Other Current Assets which, according to S&P Capital IQ's definition, includes derivative receivables, assets in commodity trading accounts, guarantee deposits, refundable deposits, bank deposits used as collateral, and marketable securities pledged as collateral. Information collected from firms' Annual Reports for 2022 validates our choice of variables. Out of 36 companies in our sample that mention margin calls in their Annual Report, 25 explain exactly where the amounts are reported in their balance sheet. Appendix A3 shows that firms indeed report cash collateral and margin payments in the three variables composing our proxy: Other Receivables, Restricted Cash, and/or Other Current Assets.

We exclude certain components of Other Current Assets which are unrelated to margins, such as accrued interest receivable and current assets of discontinued operations. We also exclude Current Derivative Assets since those reflect the fair values of derivative instruments and hedging contracts held by the company. While fair value fluctuations are informative about a company's derivatives holdings and frequently correlate with margin deposits, they may not always accurately reflect margin calls. We do not rely on reported fair values of Derivative Assets or Derivative Liabilities because they may not capture the choice of trading venues, which can affect margin requirements, the possible use of margin optimization strategies, including the recognition of netting arrangements and margin offsets across multiple contracts (e.g. within the same exchange or with the same counterparty), or any special agreements with financial institutions to secure transactions without cash collateralization.

The equation below summarizes our method for computing a Margins Indicator variable for each firm i and each quarter t. To remove the effect of firm size, we scale this variable by Total Assets measured before the start of margin calls (at the end of 2021-Q2).

$$\textit{Margins Indicator}_{i,t} = (\textit{Other Receivables}_{i,t} + \textit{Restricted Cash}_{i,t} + \textit{Other Current Assets}_{i,t} - \textit{Accrued Interest Receivable}_{i,t} - \textit{Current Assets Of Discontinued Operations}_{i,t} - \textit{Current Derivative Assets}_{i,t}) \,/\, \textit{Total Assets}_{i,\text{2021-Q2}}$$

We are not interested in the computed levels of the Margins Indicator, but rather in its time-series changes. While the levels may contain residual information about other current assets or receivables unrelated to margins, we can be reasonably confident that, in a setting with severe market movements such as the European energy crisis, large changes in the Margins Indicator would reflect margin fluctuations. Hence, for each firm, we calculate the change in the Margins Indicator from 2021-Q2 (before the margin calls) to the quarter with the largest value of the Margins Indicator during the period of price spikes from 2021-Q3 to 2022-Q3. In this way, we aim to capture a firm's maximum growth in margin deposits during the crisis and estimate the amount of liquid funds they use to cover margin calls. In summary, for each firm i, we calculate the following proxy for the size of margin calls experienced:

$$\textit{Margin Calls Proxy}_i = \max\{\textit{Margins Indicator}_{i,t} \mid t \in \textit{[2021-Q3, 2022-Q3]}\} - \textit{Margins Indicator}_{i,\text{2021-Q2}}$$

In Appendix A4 we use one of the companies in our sample as an example to illustrate the Margin Calls Proxy calculation.

Our proxy should capture collateral payments regardless of whether they originate from ETD or OTC exposures. In fact, Appendix A3 provides several examples of firms where the proxy captures collateral payments originating from OTC contracts. In the past decade, derivatives market reforms aimed at reducing systemic risks have tightened the rules for OTC trading and introduced mandatory margining similar to ETD contracts. Many institutions fall under the regulatory obligation, and even for firms that do not, it is common to voluntarily exchange collateral to reduce counterparty risk. Appendix A5 discusses in more detail the current institutional setting and regulations regarding the collateral requirements in ETD versus OTC derivatives. The bottom line is that most derivatives contracts include an exchange of collateral, although the size and form of the collateral (cash versus securities or guarantees) may be more flexible in OTC contracts compared to ETD. Our research question remains relevant as long as the majority of hedging transactions involve collateral payments. Although our study may not be relevant to cases where two firms not falling under the current regulatory obligations agree not to exchange any collateral in a bilateral OTC contract, such cases do not appear to be the norm. We believe that our analysis will continue to have important implications in a setting where the trend is for policymakers to impose stricter reforms on derivatives market participants and to require obligatory exchange of margins in an increasing number of cases.

Another consideration regarding the construction of the Margin Calls Proxy is that some energy companies may hedge both input and output prices. For example, power generating firms can enter long positions to fix purchase prices of commodities such as fuel needed for their production and short positions in power derivatives to fix the price of their future output. When energy prices rise, these firms will experience mark-to-market gains on their long positions and losses on their short positions. However, these effects will not necessarily offset each other in terms of liquidity stress. If using ETD or OTC derivatives cleared through a central clearing counterparty (CCP), they will need to deposit cash to cover margin calls on their losing positions but will not receive the on-paper gains from their winning positions because the cash deposited by their counterparties will be held by the exchange rather than transferred to their account. Therefore, this should not affect our proxy. On the other hand, if using non-centrally cleared OTC contracts, the cash collateral is bilaterally negotiated and exchanged, hence the firm may receive the cash from their long contract gains. It is unlikely that the cash received perfectly offsets the cash paid, since the firm likely enters into long and short positions with different counterparties, and there may be mismatches in the timing of payments. Nevertheless, it could alleviate at least part of the liquidity stress. Unfortunately, we are not able to reliably measure margins received, and we admit this is a limitation of our Margin Calls Proxy for power producers.[14] Even though we only take into account the margins paid, we believe that our proxy remains meaningful, and we find that power producers that possibly hold long derivatives positions are also negatively affected by the liquidity shock of margin calls.[15]

[14] As reported by S&P Capital IQ, Other Current Liabilities (which includes Other Payables) contains many variables that we are not able to exclude, introducing noise if incorporated into the proxy.

[15] Replication results of our main analyses in this subsample are available upon request.

While our Margin Calls Proxy may not be a perfect measure of margin calls size, we believe that it serves as a useful approximation, and in the rest of this section we present further evidence consistent with the proposed interpretation.

First, in Fig. 2 we plot the time-series variation in the amount of margin deposits proxied by the Margins Indicator over the sample period from 2020-Q4 to 2023-Q4. Panel A plots the unscaled Margins Indicator in millions of euros, while Panel B plots the Margins Indicator scaled by Total Assets measured before the start of margin calls (at the end of 2021-Q2) to remove the effect of firm size. The graphs show a quarterly comparison of the average value of the Margins Indicator in three terciles of companies based on the size of margin calls experienced as proxied by the Margin Calls Proxy. The blue line represents the bottom tercile of firms that experienced the smallest margin calls; the orange line is for the middle tercile; and the green line represents the top tercile of firms that experienced the largest margin calls. We can clearly see that margin deposits of companies that experience the most severe margin calls start to increase from 2021-Q3, subsequently reach their maximum value between 2022-Q1 and 2022-Q3, and thereafter decrease when market prices start to subside in the following quarters. The fact that margin deposits remain elevated up to 2022-Q3 means that companies maintained their derivatives positions instead of exiting unprofitable hedges. Therefore, we can argue that any subsequent effects on companies' financial health are unlikely to be driven by reduced hedge ratios.

Next, in Table 1 we report cross-sectional summary statistics of variables related to hedging policies and exposure to margin calls. We report statistics in the full sample, and we also conduct tests for the difference in means between firms that experience large margin calls (with Margin Calls Proxy above the median) and other firms (below the median) matched by firm size. We can see that, for the average firm in our sample, the Margin Calls Proxy is equal to 0.11. Considering that our proxy is an estimation of margin deposits as a fraction of total assets, an increase of 11 percentage points is economically significant, corresponding to approximately 2.3 billion euros of margin calls for the average firm over the period of the energy market crisis. The median is considerably smaller at 0.03, indicating a skewed distribution, with the most severe margin calls concentrated in the top quarter of companies in our sample.

To further support the validity of our Margin Calls Proxy, we develop a procedure which conducts an automated analysis and marks whether or not a firm's Annual Report for 2022 mentions any information related to margin calls (details about the procedure are available in Appendix A2). Table 1 shows that the algorithm has detected mentions of margin calls in 35 % of the reports. Firms whose Margin Calls Proxy is above the median are 27 % more likely to report experiencing margin calls in 2022 compared to firms with Margin Calls Proxy below the median matched by size.

We also report several dummy variables indicating the presence or absence in the Annual Report of keywords related to hedging, derivatives, futures, forwards, options, and swaps. In the full sample, 96 % of companies mention hedging in the Annual Report, and 98 % mention the use of derivatives. This indicates that almost all energy companies hedge price risks using derivatives. The only statistically significant difference in the propensity to mention derivatives-related keywords of firms with a Margin Calls Proxy above the median versus those below the median is in the frequency with which futures use is mentioned – firms which experience large margin calls are 25 % more likely to report the use of futures contracts. This suggests that, although all firms are likely to hedge their commodity risk exposure, some use instruments which may be subject to stricter margin requirements. This provides assurance that our subsequent results are not simply driven by the presence or absence of hedging; instead, we aim to capture the presence and size of margin calls.

To provide further evidence that our results are not driven by differences in hedging intensity, Table 1 shows statistics for several additional variables. We report statistics for the total fair value of Derivatives Positions for those firms that report them at the end of 2021-Q2 (before the margin calls), scaled by Total Assets. On average, derivatives contracts held by energy companies are worth 8.3 % of Total Assets, and about 80 % of those are current derivatives that expire within one year. While there is cross-sectional variation indicating some heterogeneity in hedging intensity, it is important to note that the mean value of derivatives holdings does not significantly differ between firms that later experience large margin calls and their peers. This means that firms above and below the median in terms of Margin Calls Proxy use derivatives contracts in equal amounts on average, but likely employ different hedging strategies, trade on different derivatives exchanges, and negotiate different contractual agreements with counterparties. All of these distinctions result in different exposures to margin call risk.

Total derivatives holdings may capture hedging of risks other than commodity price risk, such as interest rate or exchange rate risk. Next, we zoom in on firms' commodity hedging specifically. While hedge ratios are undisclosed in most cases, we can estimate them indirectly by computing each firm's risk exposure to commodity price movements. To do this, we employ an augmented market model, following a similar method to the one outlined in Carter et al. (2017). For each company in our sample, we perform a time-series regression of its stock returns on the returns of each commodity, in addition to the returns of the stock market portfolio, as illustrated by the model below:

$$R_{i,t} = \alpha_i + \beta_{m,i}R_{m,t} + \beta_{c,i}R_{c,t} + \varepsilon_{i,t},$$

where $R_{i,t}$ is the stock return of firm i in day t, $R_{m,t}$ is the return on the market portfolio (we use the STOXX Europe 600 Equal Weight as the representative index for the European stock market), $R_{c,t}$ is the return on each commodity ($R_{gas,t}$, $R_{power,t}$, and $R_{oil,t}$), and $\varepsilon_{i,t}$ is the idiosyncratic error term. If the betas $\beta_{gas,i}$, $\beta_{power,i}$ and $\beta_{oil,i}$ are different from zero, this means that firm i is exposed to commodity price risk. For a firm that outputs a given commodity, we would expect its beta to be positive since the firm's cash flows would increase when the commodity price increases, and vice versa. To hedge the risk of falling prices, the firm can use derivatives. In a perfect world, a firm that consistently hedges 100 % of its price risk would have commodity betas equal to zero. In reality, it is not uncommon for firms to hedge partially, since market frictions make perfect hedging difficult, and hedging is often costly. Prior literature has shown that hedging has an economically significant effect on the sensitivity of stock prices to changes in commodity prices, and firms that hedge their commodity price risk exhibit significantly lower betas (Tufano, 1998; Haushalter, 2000; Jin and Jorion, 2006).

To obtain the beta coefficients from the above model, we perform regressions using daily returns from 1 January 2015 to 30 June 2021, before the start of commodity market stress.[16] Table 1 shows that the mean gas, power and oil betas are all positive, consistent with the fact that the producers and distributors of these energy commodities are generally exposed to some amount of price risk, though the small magnitudes suggest a high prevalence of hedging. There are no significant differences in mean ex ante betas between firms with Margin Calls Proxies above and below the median, which suggests hedge ratios are unrelated to the subsequent size of margin calls. This result reinforces

[16] Oil Price betas are obtained by regressing the returns of Brent Crude Oil futures which are the representative contracts for oil prices in Europe. For gas and power, we consider several representative contracts across Europe and take the average betas. Specifically, to obtain Gas Price betas, we regress the returns of ICE TTF futures and UK Natural Gas Futures, which have a correlation of 95% in the period we analyze. To obtain Power Price betas, we regress the returns of Dutch Power Base Load Futures and Phelix DE Base Month Futures, which have a correlation of almost 70%; and we also estimate exposure to European wholesale electricity prices averaged across countries (obtained from: https://ember-climate.org/data-catalogue/european-wholesale-electricity-price-data/).

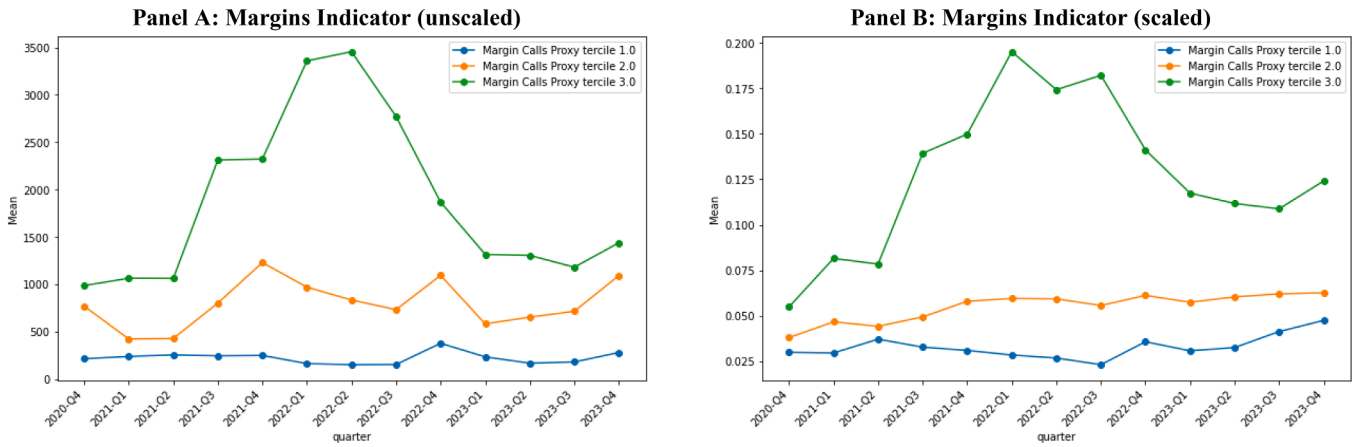


Fig. 2. Time-series variation in margin deposits by size of margin calls experienced. The following figures show the variation in margin deposits of the companies in our sample from 2020-Q4 to 2023-Q4. The amount of margin deposits is proxied by the Margins Indicator that we construct from balance sheet information (refer to Section 2.3 for details on its calculation). Panel A plots the unscaled Margins Indicator in millions of euros, while Panel B plots the Margins Indicator scaled by Total Assets measured before the start of margin calls (at the end of 2021-Q2) to remove the effect of firm size. We split companies into three terciles based on the proxied size of margin calls experienced (refer to Section 2.3 for details about our Margin Calls Proxy). The blue line plots the average value of the Margins Indicator in the bottom tercile of firms that experienced the smallest margin calls; the orange line is for the middle tercile; and the green line plots the average Margins Indicator in the top tercile of firms that experienced the largest margin calls. The plotted variables are winsorized each quarter at the 5th and 95th percentiles.

Table 1
Hedging policies and heterogeneous exposure to margin calls. This table reports cross-sectional summary statistics of variables related to hedging and margin calls. We report the number of non-missing observations, mean, standard deviation, percentiles 5, 25, 50, 75 and 95 of the distribution in the full sample. Additionally, the last two columns report the difference in means between firms with large margin calls (with Margin Calls Proxy above the median) and other firms (below the median) matched by market capitalization, and the corresponding t-statistic. The first variable is our proxy for the size of margin calls experienced by each firm (refer to Section 2.3 for details about its calculation). The second is a dummy variable equal to 1 if a firm's Annual Report for 2022 mentions any information related to margin calls, or 0 otherwise (refer to Appendix A2 for details). Next, we report several dummy variables indicating the presence or absence in the Annual Report of keywords related to hedging, derivatives, futures, forwards, options, and swaps. (Current) Derivatives Positions refers to the total fair value of (current) derivatives, for those firms that report them at the end of 2021-Q2 (before the margin calls), scaled by Total Assets. Finally, the commodity price betas are obtained by regressing each company's daily stock returns on the commodity returns from 1 January 2015 to 30 June 2021 (refer to Section 2.3 for details on the methodology).

		Full Sample Summary Statistics							Difference between firms with large margin calls and other firms matched by size	
Variable	**N**	**mean**	**std. dev.**	**p5**	**p25**	**median**	**p75**	**p95**	**mean diff.**	**t-stat**
Margin Calls Proxy	120	0.11	0.22	0.00	0.01	0.03	0.10	0.44	0.19 ***	*5.16*
Annual Report mentions margin calls	102	35 %	48 %	0 %	0 %	0 %	100 %	100 %	27 % ***	*3.50*
Annual Report mentions hedging	102	96 %	20 %	100 %	100 %	100 %	100 %	100 %	4 %	*1.00*
Annual Report mentions derivatives	102	98 %	14 %	100 %	100 %	100 %	100 %	100 %	4 %	*1.43*
Annual Report mentions futures	102	59 %	49 %	0 %	0 %	100 %	100 %	100 %	25 % **	*2.48*
Annual Report mentions forwards	102	75 %	43 %	0 %	100 %	100 %	100 %	100 %	13 %	*1.63*
Annual Report mentions options	102	70 %	46 %	0 %	0 %	100 %	100 %	100 %	10 %	*1.22*
Annual Report mentions swaps	102	82 %	38 %	0 %	100 %	100 %	100 %	100 %	8 %	*1.16*
Derivatives Positions / Total Assets	64	8.3 %	15.5 %	0.1 %	0.4 %	2.0 %	8.7 %	35.0 %	0.7 %	*0.16*
Current Derivatives Positions / Total Assets	64	6.7 %	12.6 %	0.0 %	0.2 %	1.1 %	7.0 %	31.4 %	0.1 %	*0.01*
Oil Price beta	120	0.06	0.14	−0.05	0.00	0.02	0.09	0.34	−0.03	*−1.13*
Gas Price beta	120	0.01	0.02	−0.02	−0.01	0.00	0.02	0.05	0.00	*0.04*
Power Price beta	120	0.01	0.02	−0.01	0.00	0.00	0.01	0.03	0.00	*−0.24*

the notion that our Margin Calls Proxy is capturing a distinct risk exposure to margin calls, rather than general risk exposure to commodity price movements.

Overall, Table 1 presents evidence of widespread derivatives usage, though with likely variations in risk management policies and approaches across energy firms, contributing to significant heterogeneity in firms' exposure to margin calls. Importantly, the results provide additional support for the validity of our Margin Calls Proxy.

3. Empirical analysis

In this section, we present summary statistics, we examine how companies meet margin calls, and then we proceed with our main analysis of the effects of margin calls on companies' financial health.

3.1. Cross-sectional statistics before the margin calls

We begin by examining cross-sectional differences that exist before the margin calls. Table 2 reports summary statistics of key firm characteristics, financial statement items, and financial ratios measured at the end of 2021-Q2. We show statistics in the full sample, and we also conduct tests for the difference in means between firms that subsequently experience large margin calls (with Margin Calls Proxy above the median) and other firms (below the median) matched by firm size. Once we control for firm size, there are no significant differences between the two subsamples in ex ante cash reserves, fixed assets, total debt, quick ratios, leverage ratios, CDS spreads, or profit margins. We do, however, observe that firms who later experience large margin calls have statistically higher ex ante levels of bank debt (€549 million higher, compared to a full-sample average of €996 million), term loans (€595

Table 2
Key financial metrics before the margin calls. This table reports cross-sectional summary statistics of key firm characteristics, financial statement items, and financial ratios. All variables are measured before the margin calls, at the end of 2021-Q2. We report the number of non-missing observations, mean, standard deviation, percentiles 5, 25, 50, 75 and 95 of the distribution in the full sample. Additionally, the last two columns report the difference in means between firms with large margin calls (with Margin Calls Proxy above the median) and other firms (below the median) matched by market capitalization, and the corresponding t-statistic. Fixed Assets refer to Net Property, Plant, and Equipment (PP&E). Short-Term Liabilities are equal to Current Liabilities excluding Derivatives Liabilities. Leverage Ratio is calculated as Total Liabilities/Total Assets. Cost of Debt is calculated as a weighted average of the yields on bonds issued by a company. Finally, Profit Margin (before tax) is equal to Earnings Before Tax (EBT)/Total Revenue.

		Full Sample Summary Statistics							Difference between firms with large margin calls and other firms matched by size	
Variable	**N**	**mean**	**std. dev.**	**p5**	**p25**	**median**	**p75**	**p95**	**mean diff.**	**t-stat**
(€ mill.)										
Market Capitalization	116	8704	20,163	41	250	1635	5097	43,189	192	*0.32*
Total Assets	120	21,458	52,048	41	468	2651	12,880	120,465	3617	*0.89*
Cash & Equivalents	120	1484	4545	1.6	26	177	817	7109	9.0	*0.02*
Fixed Assets	118	9821	23,999	5.4	130	1124	6399	59,374	−468	*−0.27*
Total Liabilities	120	14,436	34,266	13	207	1919	8395	79,416	3646	*1.22*
Total Debt	120	6044	14,268	0.2	55	965	4352	38,853	−278	*−0.30*
Short-Term Liabilities	120	4812	11,858	5.8	75	377	2624	28,270	1545	*1.47*
Long-Term Debt	111	4782	10,411	0.0	69	914	3667	29,571	−228	*−0.32*
Bank Debt	103	996	1606	0.0	54	432	1166	4087	549 *	*1.82*
Term Loans	95	969	1555	0.1	72	374	1210	3699	595 **	*2.33*
Drawn Revolving Credit	67	157	354	0.0	0.0	5.5	107	890	−38	*−0.43*
Undrawn Revolving Credit	52	1399	2181	21	89	396	1829	5447	1570 ***	*3.04*
Quick Ratio	120	1.99	5.33	0.29	0.67	0.91	1.34	4.66	−1.12	*−1.09*
Leverage Ratio	120	0.58	0.21	0.13	0.47	0.63	0.73	0.84	0.05	*1.14*
Cost of Debt	118	0.006	0.009	−0.002	0.001	0.004	0.008	0.017	−0.004 **	*−2.43*
5-year CDS spread, daily average (bps)	21	47.6	18.1	25.8	35.5	44.6	59.4	67.2	7.54	*0.96*
5-year CDS spread, maximum (bps)	21	51.9	19.4	28.8	40.8	46.8	62.0	73.7	7.46	*0.88*
Profit Margin (before tax)	111	−1.41	16.63	−0.27	0.02	0.10	0.22	0.56	−3.36	*−1.00*

million higher, compared to a full-sample average of €969 million), and undrawn revolving credit (€1.6 billion higher, compared to a full-sample average of €1.4 billion across 52 firms with non-missing observations for this variable). On the one hand, this suggests they tend to have preexisting relationships with lenders which may help them to weather liquidity shocks by drawing from available credit lines. On the other hand, if their revolving credit is not sufficient to cover the extreme collateral demands during the crisis, their higher amounts of existing credit may make further borrowing more difficult or costly. This half of the sample, before experiencing margin calls, exhibits an average cost of debt which is 0.4 % lower than that of their peers. This difference is statistically significant (t-statistic = −2.43) and economically significant when compared to the full-sample average of 0.6 %. We obtain each company's cost of debt from Bloomberg where it is calculated as the weighted average of the yields on bonds issued by the company.

These cross-sectional differences observed before the energy crisis may affect subsequent empirical analyses if leverage is positively correlated with hedging. It may be the case that more indebted companies, facing higher risk of bankruptcy, hedge more to reduce the risk. We verify that this is not the case in Appendix A6. The theory and evidence on the determinants of hedging are mixed, and the results in our sample are consistent with papers such as Nance, Smith, Smithson (1993) and Mian (1996) who find no significant relation between leverage and hedging. The former point out that "the unconditional impact of leverage on hedging activity is indeterminate: Greater leverage implies more hedging to control the underinvestment problem; but, since greater leverage is associated with fewer investment options, greater leverage implies less hedging." We report *t*-tests for differences in means of variables related to hedging intensity between firms with high leverage and other firms matched by size. Regardless of how we measure leverage (by total debt, long-term debt, bank debt, term loans, or undrawn revolving credit), the results reveal no significant differences in derivatives positions or commodity betas of more versus less indebted firms. This evidence helps to alleviate concerns of leverage acting as a confounding factor in our tests.

3.2. How did companies cover margin calls?

When a company experiences a large margin call, especially in the billion-euro range as was the case in the 2022 energy crisis, it produces an immense liquidity shock for the firm. The necessary cash collateral has to be delivered within a short window of time, typically under five days but sometimes as short as 24 hours. To come up with the liquid funds to cover the margin call, the company can either use up its internal cash resources, sell some of its assets, or borrow funds from third parties. In this section, we investigate how firms deal with margin calls.

As a preliminary examination of the time-series variation in key financial variables, Appendix A7 displays plots similar to Fig. 2, comparing terciles of firms sorted by the size of margin calls experienced. Panels C, E, F and G reveal pronounced jumps in the short-term borrowings, bank debt, term loans and drawn revolving credit of companies suffering large margin calls during the energy crisis. These jumps precisely match the timing of the energy price spikes: they peak most significantly in 2022-Q1 and 2022-Q3 and decrease afterwards. This suggests an important role for short-term loans and credit lines to cover immediate liquidity needs.

We test the statistical significance of these preliminary results in Table 3. We use a Difference-in-Difference (DiD) regression framework examining how companies' balance sheets change during the period from 2021-Q3 to 2022-Q3, depending on the size of margin calls they experienced, compared to the pre-shock period (2020-Q4 to 2021-Q2). The independent variables are interactions between our Margin Calls Proxy and dummy variables for each quarter during the margin calls period. The dependent variables in columns (1)-(8) are financial statement variables which shed light on whether firms facing margin calls relied on internal cash, asset sales, or various forms of credit to meet liquidity requirements. All regressions include firm and quarter fixed effects.

First, Panel A shows the effects in the full sample. The regression in column (1) shows no significant decrease in cash holdings associated with the margin calls, suggesting that firms were unlikely to be able to simply rely on their internal cash reserves to cover the sizable margin requirements. Similarly, in column (2) there is no significant decrease in

Table 3

How did companies cover margin calls?. This table presents results of Difference-in-Difference regression analyses examining how companies' balance sheets changed during the period from 2021-Q3 to 2022-Q3, depending on the size of margin calls they experienced, compared to the pre-shock period (2020-Q4 to 2021-Q2). The independent variables are interactions between our Margin Calls Proxy and dummy variables for each quarter during the margin calls period. The dependent variables in columns (1)-(8) are financial statement variables which shed light on whether firms facing margin calls relied on internal cash, asset sales, or various forms of credit to meet liquidity requirements. Fixed Assets refer to Net Property, Plant, and Equipment (PP&E). Short-Term Liabilities are Current Liabilities excluding Derivatives Liabilities. We add firm and quarter fixed effects. Panel A shows the effects in the full sample. Panel B splits the sample by the median market capitalization measured before the margin calls (in 2021-Q2). Finally, Panel C splits the sample by the median size of credit lines available before the margin calls (Undrawn Revolving Credit in 2021-Q2). All dependent variables are winsorized each quarter at the 5th and 95th percentiles. The table reports regression coefficients and the corresponding t-statistics below them in parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Panel A: Full sample

Dependent variable:	Cash & Equivalents 1	Fixed Assets 2	Short-Term Liabilities 3	Long-Term Debt 4	Bank Debt 5	Term Loans 6	Drawn Revolving Credit 7	Undrawn Revolving Credit 8
Intercept	−236.2	−542.1	−4545.0***	−234.6	−248.2	−196.4	−510.5	11.2
	(−1.10)	(−1.45)	(−2.72)	(−1.03)	(−0.83)	(−0.83)	(−1.40)	(0.01)
Margin Calls Proxy x 2021-Q3	−366.7	362.4	7779.9***	350.1	643.3	405.9	239.0	910.8
	(−1.29)	(0.73)	(3.52)	(1.26)	(1.46)	(1.17)	(0.47)	(0.59)
Margin Calls Proxy x 2021-Q4	140.5	−291.2	2799.8	566.9**	1018.8***	807.5***	876.7*	327.8
	(0.50)	(−0.59)	(1.28)	(2.06)	(3.30)	(3.31)	(1.91)	(0.13)
Margin Calls Proxy x 2022-Q1	501.3*	321.4	13,805.8***	1179.5***	1164.8***	1137.5***	1431.1***	−3336.7
	(1.78)	(0.65)	(6.29)	(4.21)	(3.38)	(4.19)	(2.77)	(−1.49)
Margin Calls Proxy x 2022-Q2	−36.3	579.5	9107.4***	1696.0***	958.2***	736.8***	250.4	1829.2
	(−0.13)	(1.18)	(4.15)	(6.07)	(3.10)	(3.01)	(0.64)	(1.30)
Margin Calls Proxy x 2022-Q3	−262.9	1056.5**	14,049.1***	1944.7***	564.5	564.0	852.3	1726.4
	(−0.93)	(2.14)	(6.41)	(7.03)	(1.26)	(1.60)	(1.37)	(1.08)
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Quarter FE	YES	YES	YES	YES	YES	YES	YES	YES
N observations	934	928	943	876	652	608	435	293
Adjusted R^2	0.915	0.996	0.826	0.995	0.880	0.883	0.647	0.872

Panel B: By firm size

Large firms

Dependent variable:	Cash & Equivalents 1	Fixed Assets 2	Short-Term Liabilities 3	Long-Term Debt 4	Bank Debt 5	Term Loans 6	Drawn Revolving Credit 7	Undrawn Revolving Credit 8
Intercept	581.1**	2780.2***	−527.7	1711.0***	1809.0***	−24.7	1629.8***	1092.4**
	(2.00)	(5.46)	(−0.23)	(6.26)	(6.29)	(−0.11)	(11.36)	(2.04)
Margin Calls Proxy x 2021-Q3	−551.3	346.2	9139.6**	394.6	770.7	524.2	536.9	1736.7
	(−1.21)	(0.43)	(2.57)	(0.92)	(1.24)	(1.05)	(0.63)	(0.69)
Margin Calls Proxy x 2021-Q4	−5.331	−635.6	3069.2	585.9	1213.6***	971.5***	1660.9**	1692.2
	(−0.01)	(−0.80)	(0.87)	(1.37)	(2.63)	(2.65)	(2.00)	(0.48)
Margin Calls Proxy x 2022-Q1	346.7	134.0	16,869.7***	1209.9***	1209.6**	1208.1***	2488.9***	−3304.8
	(0.76)	(0.17)	(4.75)	(2.82)	(2.53)	(3.16)	(2.88)	(−1.06)
Margin Calls Proxy x 2022-Q2	−310.7	570.2	11,495.8***	1810.0***	1099.9**	825.0**	847.5	4001.3
	(−0.68)	(0.72)	(3.24)	(4.23)	(2.38)	(2.24)	(1.04)	(1.48)
Margin Calls Proxy x 2022-Q3	−927.8**	975.0	16,226.3***	2054.2***	299.9	448.8	1107.5	2489.7
	(−2.04)	(1.22)	(4.58)	(4.80)	(0.48)	(0.90)	(1.28)	(0.98)
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Quarter FE	YES	YES	YES	YES	YES	YES	YES	YES
N observations	454	456	455	460	355	334	260	188
Adjusted R^2	0.901	0.995	0.814	0.994	0.843	0.831	0.628	0.845

Small firms

Dependent variable:	Cash & Equivalents 1	Fixed Assets 2	Short-Term Liabilities 3	Long-Term Debt 4	Bank Debt 5	Term Loans 6	Drawn Revolving Credit 7	Undrawn Revolving Credit 8
Intercept	−32.5	45.6	−2.062	38.4	−19.5	−3.199	−83.8	11.2
	(−1.06)	(1.64)	(−0.04)	(0.75)	(−0.39)	(−0.07)	(−0.86)	(0.13)
Margin Calls Proxy x 2021-Q3	−12.3	−30.7	55.8	−125.3	−11.2	−13.2	3.860	−8.996
	(−0.15)	(−0.42)	(0.39)	(−0.82)	(−0.06)	(−0.08)	(0.02)	(−0.04)
Margin Calls Proxy x 2021-Q4	−19.2	−100.9	84.1	−141.4	23.0	−20.4	174.6	2494.2**

(*continued on next page*)

Table 3 (*continued*)

Small firms								
Dependent variable:	Cash & Equivalents 1	Fixed Assets 2	Short-Term Liabilities 3	Long-Term Debt 4	Bank Debt 5	Term Loans 6	Drawn Revolving Credit 7	Undrawn Revolving Credit 8
	(−0.25)	(−1.43)	(0.60)	(−0.98)	(0.19)	(−0.18)	(1.18)	(2.25)
Margin Calls Proxy x 2022-Q1	103.1	−137.1*	64.1	−123.9	95.6	82.9	47.4	251.2
	(1.33)	(−1.94)	(0.46)	(−0.74)	(0.51)	(0.52)	(0.26)	(0.36)
Margin Calls Proxy x 2022-Q2	139.1*	−159.9**	−40.7	−58.5	25.6	−1.551	345.9*	196.5
	(1.79)	(−2.27)	(−0.29)	(−0.35)	(0.21)	(−0.01)	(1.71)	(0.37)
Margin Calls Proxy x 2022-Q3	228.0***	−229.6***	−39.9	−61.3	307.8	291.3	908.7**	552.6
	(2.84)	(−3.15)	(−0.29)	(−0.41)	(1.40)	(1.52)	(2.12)	(0.89)
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Quarter FE	YES	YES	YES	YES	YES	YES	YES	YES
N observations	448	440	456	384	274	251	168	102
Adjusted R^2	0.849	0.994	0.918	0.955	0.876	0.873	0.846	0.982
Panel C: By availability of credit lines								
Firms with large amounts of revolving credit available								
Dependent variable:	Cash & Equivalents 1	Fixed Assets 2	Short-Term Liabilities 3	Long-Term Debt 4	Bank Debt 5	Term Loans 6	Drawn Revolving Credit 7	Undrawn Revolving Credit 8
Intercept	324.9	1147.7**	−1365.9	1481.8***	4.609	−30.3	132.6	3752.5***
	(0.98)	(2.00)	(−0.58)	(4.25)	(0.01)	(−0.12)	(0.56)	(6.32)
Margin Calls Proxy x 2021-Q3	826.4	540.2	20,792.2**	486.4	−883.8	−3432.2**	712.5	−2522.6
	(0.65)	(0.25)	(2.30)	(0.37)	(−0.50)	(−2.53)	(0.69)	(−0.76)
Margin Calls Proxy x 2021-Q4	2233.2*	1007.7	4825.2	1748.7	1038.5	−104.9	1249.6	−1928.2
	(1.77)	(0.46)	(0.53)	(1.31)	(0.89)	(−0.12)	(1.22)	(−0.45)
Margin Calls Proxy x 2022-Q1	4633.0***	1253.1	31,176.7***	2853.1**	1710.4	−1096.2	2881.8***	−4502.0
	(3.67)	(0.57)	(3.45)	(2.14)	(0.95)	(−0.80)	(2.78)	(−1.35)
Margin Calls Proxy x 2022-Q2	1766.3	−1012.4	14,673.9	2686.9**	644.1	315.9	393.4	1780.1
	(1.40)	(−0.46)	(1.62)	(2.02)	(0.56)	(0.37)	(0.59)	(0.89)
Margin Calls Proxy x 2022-Q3	2396.9*	229.7	41,414.8***	2969.5**	3962.1**	1986.1	1527.8	5372.5
	(1.88)	(0.10)	(4.59)	(2.23)	(2.19)	(1.46)	(1.46)	(1.63)
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Quarter FE	YES	YES	YES	YES	YES	YES	YES	YES
N observations	203	204	207	208	171	158	146	133
Adjusted R^2	0.872	0.993	0.806	0.988	0.876	0.875	0.432	0.765
Firms with small amounts of revolving credit available								
Dependent variable:	Cash & Equivalents 1	Fixed Assets 2	Short-Term Liabilities 3	Long-Term Debt 4	Bank Debt 5	Term Loans 6	Drawn Revolving Credit 7	Undrawn Revolving Credit 8
Intercept	1056.6***	3584.0***	2923.4***	1716.1***	1821.3***	−31.3	1588.6***	116.2***
	(19.86)	(59.09)	(22.49)	(27.29)	(28.11)	(−0.54)	(29.32)	(2.76)
Margin Calls Proxy x 2021-Q3	18.4	−35.5	−51.9	−94.9	−21.2	21.0	−59.4	49.4
	(0.06)	(−0.11)	(−0.07)	(−0.25)	(−0.05)	(0.05)	(−0.17)	(0.28)
Margin Calls Proxy x 2021-Q4	−44.9	−307.5	−135.9	−82.2	−26.4	−21.8	−15.4	818.8
	(−0.16)	(−0.95)	(−0.20)	(−0.22)	(−0.07)	(−0.06)	(−0.05)	(1.33)
Margin Calls Proxy x 2022-Q1	209.0	−434.8	26.9	−187.7	−28.5	−0.994	−54.3	113.2
	(0.73)	(−1.34)	(0.04)	(−0.49)	(−0.07)	(−0.00)	(−0.15)	(0.23)
Margin Calls Proxy x 2022-Q2	442.6	−474.5	−360.4	−27.3	−31.7	−4.409	358.1	79.9
	(1.55)	(−1.46)	(−0.52)	(−0.07)	(−0.09)	(−0.01)	(0.89)	(0.22)
Margin Calls Proxy x 2022-Q3	498.1*	−634.1*	−523.9	−22.8	63.4	100.5	1442.2	454.6
	(1.73)	(−1.94)	(−0.75)	(−0.06)	(0.15)	(0.26)	(1.53)	(0.93)
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Quarter FE	YES	YES	YES	YES	YES	YES	YES	YES
N observations	206	206	206	194	162	145	150	116
Adjusted R^2	0.823	0.988	0.875	0.982	0.970	0.977	0.897	0.778

fixed assets, implying that the majority of firms have not had to resort to selling assets in order to meet liquidity demands. Next, to examine companies' short-term debt, we focus on current liabilities, from which we exclude any current derivative liabilities. Consistent with the preliminary evidence from the plots, the regression in column (3) reveals a positive and significant association between margin calls size and increase in short-term liabilities. Coefficients are largest and most significant on the interactions of Margin Calls Proxy with the dummies for 2022-Q1 (t-statistic = 6.29) and 2022-Q3 (t-statistic = 6.41) – the two quarters with the largest jumps in energy prices. Columns (5)-(7) reveal significant increases in bank debt, term loans, and drawn revolving credit for firms that experience margin calls, with the most significant effects in 2022-Q1 (t-statistics = 3.38, 4.19, and 2.77, respectively). At the same time, undrawn revolving credit seems to drop during 2022-Q1, consistent with firms using their credit lines. Although the coefficient in column (8) is not statistically significant (t-statistics = −1.49), this could be due to the low number of non-missing observations for this variable. (Revolving credit amounts are available in S&P Capital IQ for approximately half of the firms in our sample.) Interestingly, the regression in column (4) reveals a rise in long-term debt as well for firms affected by margin calls. The effect grows over the quarters and is most significant in 2022-Q3 (t-statistic = 7.03). This indicates that perhaps firms used their credit lines and short-term borrowings in the first wave of margin calls to obtain liquid funds quickly, while during 2022 they were able to obtain long-term debt to prepare for the second wave. Overall, the results in Panel A of Table 3 show that the majority of firms facing large margin calls rely on various forms of credit to satisfy cash collateral deposits.

Next, in Panel B, we investigate whether large and small firms deal differently with margin calls. Smaller firms are less likely to possess large amounts of cash reserves and may not have existing relationships with lenders. For this reason, they may be forced to sell assets in order to obtain liquid funds to cover margin requirements. To test this conjecture, we split the sample by the median market capitalization measured before the margin calls (in 2021-Q2), and we perform the same regressions as in Panel A in the two subsamples separately.

The results in the subsample of large firms mirror those in the full sample. The only difference is in the regression of cash and equivalents in column (1) where we observe a negative and significant coefficient on the interaction of Margin Calls Proxy and the dummy for 2022-Q3 (t-statistic = −2.04). This implies that large firms, which are more likely to have liquidity buffers, used their cash reserves for margin deposits, and this led to a marked reduction in their cash holdings during the most severe wave of margin calls.

On the other hand, the results in the subsample of small firms differ. The regression in column (2) reveals a significant negative relationship between margin calls size and change in fixed assets during 2022 (t-statistic = −3.15 on the interaction with 2022-Q3). This suggests that small firms, which may have relatively smaller cash reserves and reduced ability to borrow, resort to selling assets in order to gather liquid funds to cover growing margin requirements. The results in columns (3)-(6) are all insignificant, which means small firms were not able to obtain loans to deal with the liquidity shock. The regressions in columns (7)-(8) reveal an increase in undrawn revolving credit in 2021-Q4 for small firms that were more affected by margin calls (t-statistic = 2.25), and subsequent increases in drawn revolving credit especially in 2022-Q3 (t-statistic = 2.12). Unlike large firms which likely had access to revolving credit in advance, it seems that perhaps small firms had to apply for credit lines at the first signs of market turmoil, and at least some of them were able to use this form of credit to cover the most severe margin calls.

Furthermore, in Panel C, we study how firms dealt with margin calls depending on their ex ante availability of credit lines. We split the sample by the median size of credit lines available before the margin calls (Undrawn Revolving Credit in 2021-Q2), and we repeat the regressions in these two subsamples separately. While these results are informative, they do not cover all firms because the variable Undrawn Revolving Credit has non-missing observations only for about half of the firms in our sample.

The results in the subsample of firms with large amounts of revolving credit available mostly mirror those in the full sample. The only difference is that their amount of term loans does not seem to increase due to the margin calls, which makes sense if they are able to rely on their credit lines.

In contrast, in the subsample of firms with small amounts of revolving credit available ex ante, we observe a significant result only in column (2): the coefficient for fixed assets in 2022-Q3 is negative (t-statistic = −1.94). It appears that firms which were unable to use credit lines were also unable to obtain other forms of credit to cover the liquidity requirements and were ultimately forced to liquidate or divest assets to cover margin calls.

Overall, the results presented in this section reveal the various ways in which different companies handle growing margin requirements. Although borrowing seems to be the preferred source of liquid funds for most firms, we can see that this is not a viable option for all. Firms unable to obtain credit must resort to asset sales, which in times of market stress may be done at unfavorable prices, leading to losses. Regardless of which method firms use to cover margin calls, it has an impact on their balance sheet, with some firms becoming more indebted and others reducing their asset base. In the following sections, we investigate the lasting effects of these changes on firms' financial health and future outcomes.

3.3. Margin calls and default risk

When companies experience a significant build-up of losses on derivative contracts triggering liquidity shocks from margin calls with magnitudes in billions of euros, we can expect that their probability of default will increase. In this subsection, we investigate the immediate and long-term effects of margin calls on default risk. To do this, we study Credit Default Swaps (CDS) spreads, which reflect the market's perception of the likelihood that a firm will default on its debt obligations. A higher CDS spread indicates lower creditworthiness or greater financial stress. We take the spreads of 5-year CDS contracts which tend to be the most liquid ones. CDS spreads are available for a subsample of 21 companies that have CDS contracts issued on their debt.

Starting with an examination of how CDS spreads evolve during the energy crisis, Panel I in Appendix A7 plots: i) the average daily CDS spread, and ii) the maximum daily CDS spread in each quarter, for each tercile of firms sorted by the Margin Calls Proxy. Up to the end of 2021, CDS spreads of all terciles are stable and co-move together. However, starting from 2022-Q1, the average and maximum CDS spread of firms in the top tercile by margin calls size increase sharply and surpass those of their peers. They continue to grow together with energy prices over the next two quarters, peaking in 2022-Q3 at an average (maximum) value that is 3 (3.6) times higher than their pre-margin calls level, and is substantially larger than the increase experienced by other companies. Although CDS spreads decline after the market turmoil, the spreads of the firms that experienced the largest margin calls remain elevated until the end of our sample. They never return to their pre-margin calls levels and remain larger than the CDS spreads of firms in the bottom tercile. Overall, this preliminary evidence points to a significant potential impact of margin calls on firms' credit risk and probability of default.

To test the significance of these effects, in Table 4 we perform Difference-in-Difference (DiD) analyses examining the changes in CDS spreads that companies experience after the energy market turmoil, depending on the size of margin calls they suffer. The dependent variable is the daily CDS spread, expressed in basis points, during different intervals from 10 days to almost 1 year after the margin calls. The independent variable of interest is an interaction between our Margin Calls Proxy and a Post Margin Calls dummy variable which is equal to one after the largest spike in energy prices on 26 August 2022. Thus, to examine how the magnitude of margin calls experienced by different

Table 4

Margin calls and default risk. This table reports the results of Difference-in-Difference analyses examining the changes in default risk that companies experience after the market turmoil, depending on the size of margin calls they suffered. The dependent variable is the 5-year Credit Default Swap (CDS) daily spread in basis points during different intervals, from 10 days to almost 1 year after the margin calls. The independent variable is an interaction between our Margin Calls Proxy and a Post Margin Calls dummy variable which is equal to 1 after the largest jump in energy prices on 26 August 2022. The full sample period extends from 30 December 2020 to 30 December 2023. Panel A compares CDS spreads after 26 August 2022 to spreads before that date, while Panel B excludes the quarters of market turmoil, thus comparing CDS spreads after 26 August 2022 to spreads before 1 July 2021. These analyses restrict the sample to 21 companies that have CDS contracts issued on their debt. For each firm, we require at least half of the observations in each pre- and post-shock period to be non-missing. We include firm and day fixed effects. The table reports regression coefficients and the corresponding t-statistics below them in parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Panel A: Full Sample Period (30 Dec 2020 - 30 Dec 2023)								
Dependent variable: Daily CDS spread (bps)	Number of Trading Days Post Margin Calls							
	(10 days)	(21 days)	(42 days)	(63 days)	(126 days)	(189 days)	(252 days)	(351 days)
Intercept	40.84***	40.76***	40.51***	41.04***	44.13***	45.57***	46.84***	48.04***
	(9.20)	(8.98)	(8.94)	(8.97)	(9.88)	(10.57)	(11.26)	(12.08)
Margin Calls Proxy	23.28***	23.65***	22.83***	19.86***	12.12***	8.70***	5.69***	2.93**
x Post Margin Calls	(3.81)	(5.90)	(7.87)	(8.13)	(6.74)	(5.81)	(4.33)	(2.56)
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES	YES	YES	YES
N observations	6615	6804	7119	7434	8379	9324	10,269	11,760
Adjusted R^2	0.712	0.713	0.726	0.732	0.732	0.731	0.732	0.731
Panel B: Excluding Margin Calls Period (excl. 1 Jul 2021 - 26 Aug 2022)								
Dependent variable: Daily CDS spread (bps)	Number of Trading Days Post Margin Calls							
	(10 days)	(21 days)	(42 days)	(63 days)	(126 days)	(189 days)	(252 days)	(351 days)
Intercept	36.55***	36.84***	36.90***	38.94***	46.87***	49.61***	51.61***	53.04***
	(12.53)	(9.03)	(8.66)	(8.81)	(11.49)	(13.37)	(15.10)	(17.10)
Margin Calls Proxy	34.58***	34.95***	34.13***	31.17***	23.42***	20.01***	17.00***	14.23***
x Post Margin Calls	(8.98)	(9.73)	(11.92)	(12.02)	(11.80)	(12.01)	(11.60)	(11.17)
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES
Time FE	YES	YES	YES	YES	YES	YES	YES	YES
N observations	2016	2205	2520	2835	3780	4725	5670	7161
Adjusted R^2	0.781	0.717	0.757	0.773	0.780	0.787	0.796	0.804

firms affects their default risk post-shock, we are interested in the coefficient on this interaction term. The full sample period extends from 30 December 2020 to 30 December 2023. We include firm and time fixed effects. Panel A compares CDS spreads after 26 August 2022 to spreads before that date. The interaction coefficient is positive and highly significant in all columns, pointing to a definite increase in default risk after 26 August 2022 for firms that experience larger margin calls. The coefficients exhibit a decreasing pattern over time, ranging from 23.28 (t-statistic = 3.81) in the ten days after the date, to 2.93 (t-statistic = 2.56) in the 351 days post-margin calls. Therefore, the effect of margin calls on default probabilities is strongest in the months immediately after the shock, but it remains significant for at least a year afterwards. In terms of economic significance, the interaction coefficients imply that a one-standard-deviation increase in the size of margin calls experienced results in a 10.6 % (1.3 %) higher CDS spread in the following 10 (351) days. In Panel B, we repeat the analysis excluding the quarters of market turmoil, thus comparing CDS spreads after 26 August 2022 to spreads before 1 July 2021. Once again, the interaction coefficients are all positive and highly significant, and exhibit a similar pattern over time. The effects are larger than in Panel A, ranging from 34.58 (t-statistic = 8.98) in the ten days after 26 August 2022, to 14.23 (t-statistic = 11.17) in the 351 days after the date. The economic significance is larger too, with a one-standard-deviation increase in the size of margin calls resulting in a 15.7 % (6.5 %) higher CDS spread in the following 10 (351) days.

Clearly, experiencing a margin call increases a firm's credit risk and probability of default, as reflected by elevated CDS spreads for a long time afterwards. Next, we examine how the market's perception of elevated default risk spills over to creditors' estimations of the firm's creditworthiness.

3.4. Long-term effect of margin calls on the cost of debt

The fact that firms affected by margin calls experience a larger increase in default risk after the energy market crisis as compared to other firms leads us to examine potential long-term impacts on their borrowing costs. If the market estimates a higher risk of financial distress for these companies in the year after the margin call shocks, we may expect creditors to have reduced confidence in their creditworthiness. This may be reflected in these firms' cost of debt. To test this hypothesis, we obtain each firm's cost of debt at the end of each quarter from Bloomberg, where it is calculated as the weighted average of the yields on bonds issued by the company. These values reflect market-implied borrowing costs, measuring the return investors demand for holding the company's bonds. Higher yields indicate greater perceived credit risk and represent the cost that the company would incur if it were to issue new debt at the time of measurement.

We use a similar DiD framework as in Table 4. Our methodology relies on interacting the proxy for the size of margin calls experienced by different firms with a post-shock indicator for the period after the margin calls. The goal is to examine how the energy market crisis differentially affects firms' long-term cost of debt based on their varying levels of exposure to margin calls. We report results in Table 5. The Post Margin Calls dummy variable is equal to one after the margin calls in 2022-Q3 and in each subsequent quarter up to the end of 2023. Hence, we capture effects lasting for over a year after the shock. The quarters during the market turmoil are excluded, thus comparing the post-margin calls effect with the average cost of debt in the pre-shock period (2020-Q4 to 2021-Q2). The first column in Table 5 shows the effects in the full sample; the next two columns split the sample by the median market capitalization measured before the margin calls (in 2021-Q2); and the last two columns split the sample by the median size of credit lines available before the margin calls (Undrawn Revolving Credit in 2021-Q2). We add firm and quarter fixed effects to the regressions.

Table 5
Long-term effect of margin calls on the cost of debt. This table reports the results of Difference-in-Difference regression analyses examining how companies' borrowing costs change after the energy market crisis, depending on the size of margin calls they experienced. The dependent variable is the Cost of Debt calculated as a weighted average of the yields on bonds issued by a given company. The independent variable is an interaction between our Margin Calls Proxy and a Post Margin Calls dummy variable which is equal to 1 after the margin calls in 2022-Q3 and up to the end of 2023. The quarters of market turmoil are excluded, thus comparing the post-margin calls effect with the average cost of debt in the pre-shock period (2020-Q4 to 2021-Q2). The first column shows the effects in the full sample; the next two columns split the sample by the median market capitalization measured before the margin calls (in 2021-Q2); and the last two columns split the sample by the median size of credit lines available before the margin calls (Undrawn Revolving Credit in 2021-Q2). We add firm and quarter fixed effects. The dependent variable is winsorized each quarter at the 5th and 95th percentiles. The table reports regression coefficients and the corresponding t-statistics below them in parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Dependent variable: Cost of Debt	Sample				
	Full sample	Large firms	Small firms	Firms with large credit lines available	Firms with small credit lines available
Intercept	0.009**	−0.005*	0.003	−0.007***	−0.007**
	(2.55)	(−1.94)	(0.76)	(−2.95)	(−2.02)
Margin Calls Proxy x Post Margin Calls	0.002	−0.001	0.020***	−0.005	0.045***
	(0.86)	(−0.39)	(2.97)	(−0.78)	(3.44)
Firm FE	YES	YES	YES	YES	YES
Quarter FE	YES	YES	YES	YES	YES
N observations	870	466	376	219	203
Adjusted R^2	0.833	0.858	0.807	0.860	0.795

The interaction coefficients in the full sample and in the subsamples of large firms and firms with large credit lines available are not statistically significant, showing no effects of margin calls on the cost of debt. Although these firms' debt levels rise as a result of experiencing margin calls, their existing relationships with lenders help to supply them with the needed liquidity quickly and at pre-negotiated rates. In contrast, the coefficients in the subsamples of small firms and firms with small credit lines available are positive and significant (t-statistics = 2.97 and 3.44, respectively). Hence, firms which are less likely to have preexisting relationships with lenders or pre-approved credit lines are more susceptible to the long-term negative effects of margin calls. Those of them that experience larger margin calls face more severe increases in their borrowing costs in subsequent quarters, and this effect lasts for more than a year after the crisis. The economic significance of these results is staggering: In the subsample of small firms (firms with small credit lines available), a one-standard-deviation increase in the size of margin calls experienced results in 43 % (50 %) higher cost of debt subsequently. This impressive rise in borrowing costs reflects creditors' perception of these firms as more financially vulnerable following the liquidity shocks, rendering them as higher-risk borrowers with an uncertain ability to meet obligations in case of further market volatility.

3.5. Net effect on profit margins

Our analysis of the impacts of margin calls on company financial outcomes concludes with an examination of profit margins. Hedging with derivatives involves a trade-off of benefits and potential drawbacks. It is possible that companies recognize the risk of margin calls and take it into account when devising their risk management strategies. Perhaps, despite the liquidity pressures and heightened risk exposure, firms exposed to margin calls are able to maintain their competitive positions in the energy market and successfully preserve profitability. To explore this hypothesis, we analyze their net profit margins, which provide insight into their ability to maintain financial performance

Table 6
Net effect on profit margins. This table reports the results of Difference-in-Difference analyses examining how companies' profit margins are impacted by the energy market crisis, depending on the size of margin calls experienced. The dependent variable is Profit Margin calculated before tax as EBT/Total Revenue, to exclude any effect of windfall taxes. The independent variables are interactions of the Margin Calls Proxy with dummy variables for each quarter after the margin calls in 2022-Q3 and up to the end of 2023. The quarters of market turmoil are excluded, thus comparing the post-margin calls effect with the average profitability in the pre-shock period (2020-Q4 to 2021-Q2). The first column shows the effects in the full sample; the next two columns split the sample by the median market capitalization measured before the margin calls (in 2021-Q2); and the last two columns split the sample by the median size of credit lines available before the margin calls (Undrawn Revolving Credit in 2021-Q2). We include firm and quarter fixed effects. The dependent variable is winsorized each quarter at the 5th and 95th percentiles. The table reports regression coefficients and the corresponding t-statistics below them in parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Dependent variable: Profit Margin (before tax)	Sample				
	Full sample	Large firms	Small firms	Firms with large credit lines available	Firms with small credit lines available
Intercept	−0.058	−0.034	−0.103	0.026	−0.170**
	(−0.99)	(−0.82)	(−1.34)	(0.75)	(−2.39)
Margin Calls Proxy x 2022-Q3	−0.157**	−0.148**	−0.022	−0.054	−0.168
	(−2.12)	(−2.16)	(−0.11)	(−0.39)	(−0.42)
Margin Calls Proxy x 2022-Q4	−0.052	0.011	−0.036	0.193	−0.546
	(−0.69)	(0.15)	(−0.16)	(1.40)	(−1.01)
Margin Calls Proxy x 2023-Q1	−0.030	0.024	−0.014	0.069	−0.430
	(−0.41)	(0.35)	(−0.07)	(0.50)	(−1.08)
Margin Calls Proxy x 2023-Q2	−0.014	−0.043	0.268	−0.054	0.450
	(−0.20)	(−0.63)	(1.43)	(−0.39)	(1.13)
Margin Calls Proxy x 2023-Q3	−0.056	−0.033	−0.029	−0.046	0.544
	(−0.77)	(−0.48)	(−0.16)	(−0.33)	(1.37)
Margin Calls Proxy x 2023-Q4	−0.116	−0.108	−0.050	−0.009	0.161
	(−1.58)	(−1.58)	(−0.26)	(−0.07)	(0.40)
Firm FE	YES	YES	YES	YES	YES
Quarter FE	YES	YES	YES	YES	YES
N observations	1024	508	480	229	230
Adjusted R^2	0.518	0.614	0.476	0.759	0.411

while navigating the challenges of the energy crisis.

Following the same DiD methodology as in Table 5, we examine the differential impact of the energy market crisis on companies' profit margins, depending on the size of margin calls experienced. Results are reported in Table 6. The dependent variable is calculated as Earnings Before Tax (EBT) divided by Total Revenue. We measure profit margin before taxes to exclude any effect of windfall taxes that were imposed on energy corporations in several European countries during the sample period. The regression specifications in Table 6 are the same as in Table 5, except that instead of using a single Post Margin Calls dummy, we examine interactions of the Margin Calls Proxy with indicator variables for each quarter after 2022-Q3 and up to the end of 2023. We do this because the impact of margin calls on firm profitability seems to be relatively short-term. Although all interaction coefficients are negative in the full sample, pointing to lower profit margins for companies affected by larger margin calls, the negative effect is statistically significant only in 2022-Q3 (t-statistic = −2.12). When examining the results in the different subsamples, we observe that the effect is concentrated among large firms. Overall, the inability of affected companies to maintain the same financial performance as their peers after the largest peak in energy prices suggests that they had not anticipated the magnitude of margin calls that they experienced and were not fully prepared to deal with them in advance.

Largely, the results presented in the last three subsections reveal significant negative impacts of margin calls on companies' financial health and bottom lines. The adverse consequences for firms' default risk and cost of debt are long-lasting and are most severe for smaller firms and firms with little available credit lines. Net profit margins, especially for large firms, are negatively affected by margin calls as well. Although the drop in profitability is temporary, it points to the likelihood that firms were unprepared for the liquidity shocks that they experienced in the 2022 energy crisis. Therefore, firms may need to factor margin call risk into their risk management strategies.

3.6. Tests of the common trends assumption

Finally, in Table 7, we present tests of the common trends assumption required for the validity of the Difference-in-Differences (DiD)

Table 7

Tests of the common trends assumption. In this table, we present tests of the Difference-in-Difference common trends assumption. The regression design mirrors previous tables, except the Margin Calls Proxy is interacted with dummy variables for the quarters before the start of margin calls (the baseline quarter is 2020-Q4, and the interacted quarters are 2021-Q1 and 2021-Q2). We demonstrate common trends in CDS spreads (Panel A), Cost of Debt (Panel B), and Profit Margins (Panel C) across firms before the margin calls. CDS spreads in Panel A are those of 5-year CDS contracts and are expressed in basis points. To aggregate daily spreads by quarter, we take the average in column (1), or the maximum in column (2). The sample in this panel is reduced to 21 companies that have CDS contracts issued on their debt. In Panel B, Cost of Debt is a weighted average of the yields on bonds issued by a given firm. Finally, in Panel C, Profit Margin is calculated before tax as EBT/Total Revenue. We include firm and quarter fixed effects in all regressions. The tables report regression coefficients and the corresponding t-statistics below them in parentheses. ***, **, and * indicate statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Panel A: CDS spreads

Dependent variable:	Average daily CDS spread (bps)	Maximum daily CDS spread (bps)
	1	2
Intercept	43.1***	44.3***
	(23.89)	(28.31)
Margin Calls Proxy x 2021-Q1	2.491	0.001
	(0.79)	(0.00)
Margin Calls Proxy x 2021-Q2	4.451	2.612
	(1.41)	(0.96)
Firm FE	YES	YES
Quarter FE	YES	YES
N observations	63	63
Adjusted R^2	0.954	0.968

Panel B: Cost of Debt

Dependent variable: Cost of Debt	Sample				
	Full sample	Large firms	Small firms	Firms with large credit lines available	Firms with small credit lines available
Intercept	0.003**	−0.004***	0.003	−0.005***	−0.005***
	(2.12)	(−5.27)	(1.63)	(−5.20)	(−3.92)
Margin Calls Proxy x 2021-Q1	−0.001	0.000	−0.005	−0.002	−0.006
	(−0.76)	(0.04)	(−0.85)	(−0.58)	(−0.85)
Margin Calls Proxy x 2021-Q2	−0.000	−0.000	−0.001	−0.000	−0.004
	(−0.20)	(−0.03)	(−0.31)	(−0.13)	(−0.73)
Firm FE	YES	YES	YES	YES	YES
Quarter FE	YES	YES	YES	YES	YES
N observations	306	159	137	74	69
Adjusted R^2	0.913	0.934	0.882	0.927	0.866

Panel C: Profit Margins

Dependent variable: Profit Margin (before tax)	Sample				
	Full sample	Large firms	Small firms	Firms with large credit lines available	Firms with small credit lines available
Intercept	−0.002	−0.010	−0.013	−0.008	−0.050
	(−0.03)	(−0.15)	(−0.13)	(−0.21)	(−0.51)
Margin Calls Proxy x 2021-Q1	0.009	0.016	0.061	0.062	0.004
	(0.12)	(0.20)	(0.33)	(0.54)	(0.01)
Margin Calls Proxy x 2021-Q2	0.008	0.026	0.054	−0.117	0.132
	(0.10)	(0.32)	(0.29)	(−1.02)	(0.31)
Firm FE	YES	YES	YES	YES	YES
Quarter FE	YES	YES	YES	YES	YES
N observations	341	167	162	74	77
Adjusted R^2	0.633	0.639	0.655	0.879	0.516

model employed in previous tables. This assumption posits that, in the absence of a treatment, the outcomes for treated and control groups would have followed parallel trends over time. In the setting of our study, testing this ensures that any observed differences in outcomes post-margin calls among firms with varying degrees of exposure to the margin calls can be attributed to the shock of the margin calls rather than any preexisting trends. To show this, we perform regressions that mirror previous tables, except the Margin Calls Proxy is interacted with dummy variables for the quarters before the start of margin calls (the baseline quarter is 2020-Q4, and the interacted quarters are 2021-Q1 and 2021-Q2). Thus, the regressions test whether the pre-treatment trends in outcomes are statistically indistinguishable among firms with different sizes of margin calls. In Panels A, B and C, we demonstrate common trends in CDS spreads, cost of debt, and profit margins, respectively. To aggregate daily CDS spreads by quarter, we take the average in column (1), or the maximum in column (2) of Panel A. The dependent variables in Panels B and C are the same as in previous analyses, and we show results in the full sample as well as subsamples split by firm size and credit lines availability. Firm and quarter fixed effects are included in all regressions.

The results in the regressions in all three panels of Table 7 are statistically insignificant, thus establishing common trends in all of the outcome variables in the period before margin calls. This reinforces the reliability of our main results in earlier DiD analyses, ensuring the validity of the reported effects of margin calls on firms' financial outcomes.

4. Discussion and implications

Our results carry important managerial and regulatory implications. From a managerial perspective, devising a risk management strategy utilizing derivative instruments requires consideration of an important trade-off: It allows a corporation to reduce market price risks, but it may introduce counterparty credit risk and margin/liquidity risk.[17] Beyond conventional market risk assessments, hedgers should stress-test their exposure to potential margin shocks and maintain liquidity buffers or pre-arranged credit lines sufficient to absorb sudden collateral demands. The choice between maintaining internal liquid reserves and negotiating credit lines also involves a careful balance. On the one hand, there is an opportunity cost of accumulating cash, as these resources become unavailable for investment, debt repayment, or other uses. On the other hand, while our findings highlight the importance of establishing long-term banking relationships, relying entirely on revolving credit may turn problematic if banks themselves are under stress during extreme market volatility. Indeed, the European Banking Authority (EBA) reports in a September 2022 response to the European Commission[18] that banks have significantly increased their exposure to the energy sector as a result of the support they have provided to energy companies in managing the volatility in derivatives markets. To the extent that margin requirements are satisfied through short-term borrowing in money markets, spillovers are possible to those markets as well. In sum, severe margin calls draw regulatory attention to multiple issues: the liquidity strain on derivatives users and banks, the accumulation of risks that can occur in interrelated and concentrated markets, and the potential rise of financial instability. While central clearing and daily margining are designed to safeguard counterparties and market integrity, they may unintentionally propagate systemic liquidity shortages when volatility spikes. This underlines the need to review current margining practices. Policymakers have proposed broadening the list of eligible assets that CCPs can accept as collateral and the possible provision of bank guarantees as collateral. However, these measures are typically not favored by market participants and are implemented only temporarily. A remaining challenge is the limited data availability and the difficulty of predicting margin calls. The EBA and BIS urge for improved transparency of margin models to allow market participants to anticipate their liquidity needs better. Recent BIS reports[19] outline policy proposals and recommendations to streamline margin processes in both centrally cleared and non-centrally cleared markets. These measures could improve resilience in times of market stress without undermining the fundamental principles of collateralization.

These implications extend beyond the energy sector. The European energy crisis of 2022 may appear as an unusually severe market dislocation, but considering recent geopolitical tensions and persistent market uncertainty, similar crises could develop in the future. This makes continued careful monitoring and assessment of risks imperative for derivatives market participants and regulators.

5. Conclusion

Despite the numerous benefits of hedging documented in the risk management literature, there are potential downsides which have remained understudied. In particular, we show that companies who use derivatives to hedge price risks are exposed to liquidity risks arising from margin fluctuations, especially in times of market stress. In this study, we construct a novel proxy for margin calls using balance sheet data, and we investigate the impact of margin call shocks on companies' financial stability and performance in the short and long term.

The European energy crisis in 2022 provides an excellent empirical setting for our analysis. Commodity markets in general are very volatile and users of commodity derivatives need to post considerable cash collaterals. During 2022, oil, gas and electricity prices in Europe surged, and since energy companies tend to be on the short side of derivatives contracts, the value of their hedge positions plummeted and many experienced unprecedented margin calls. We exploit the cross-sectional heterogeneity in hedging strategies to investigate the effects of the market shock on firms with different exposures to margin calls.

We document that most companies facing substantial margin calls meet liquidity requirements by using credit in various forms, including short-term and long-term bank debt, term loans, and credit lines. Smaller firms which may have weaker relationships with lenders, as well as firms without availability of pre-approved revolving credit, resort to asset sales in order to cover margin calls. Default risk, measured by CDS spreads, increases more for firms facing larger margin calls after the most severe spike in energy prices, and remains elevated for at least one year afterwards. Moreover, we uncover significant increases in the cost of debt persisting for more than a year after the margin calls, for affected small firms and firms with restricted access to credit lines. Finally, there is a short-term drop in net profit margins for companies that experience margin calls, especially in the subsample of large firms. Although the drop in profitability is temporary, it suggests that firms lacked sufficient preparedness for the liquidity shocks they faced in the 2022 energy crisis. This event may serve as a cautionary tale for firms to consider the risk of margin calls when devising their risk management strategies.

To our knowledge, we are the first study to document the negative impacts of margin calls on companies' financial stability and performance. We show that the reported effects are economically significant and are not driven by common trends, firm characteristics, or differences in hedge ratios. Instead, they reveal a unique source of risk for firms that hedge with derivatives, which is likely to arise in times of market stress.

[17] An article by Sparla, Lehrbass and Römmich (2023) discusses this "Risk Trilemma" and provides a case study of margin risk for a stylized German power producer.

[18] https://www.eba.europa.eu/sites/default/files/document_library/About%20Us/Missions%20and%20tasks/Correspondence%20with%20EU%20institutions/2022/1039915/EBA%20response%20to%20EC%20request%20on%20energy%20markets.pdf

[19] Available at: https://www.bis.org/bcbs/publ/d590.htm and https://www.bis.org/bcbs/publ/d589.htm.

Our findings may be generalizable to any derivatives market where companies are exposed to the risk of mounting collateral demands. Overall, our study highlights the importance of managing liquidity risks associated with derivatives use. We contribute to ongoing debates about the regulation of derivatives trading, central clearing counterparties, margining rules and practices, and their implications for financial stability. Sustained risk monitoring and coordination between the corporate sector and policymakers are crucial to ensure that risk management practices continue to reinforce financial resilience.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Chat GPT in order to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Appendix

A1. Sample filters

We exclude from our sample:

i. companies with missing values for the main variables in our analyses (Ticker, Total Assets, Total Liabilities, Margins Indicator and Margin Calls Proxy) across all quarters;
ii. companies with a closing year-end stock price of <1 euro at any point during the sample;
iii. companies with negative Total Equity in the period before the margin calls; and
iv. companies with Total Liabilities close to zero (below 0.05 million euros) in any quarter during the sample - this criterion filters out a few holding companies.

A2. Procedure for marking firms that mention margin calls in the Annual Report

We develop a procedure which conducts an automated analysis and marks whether or not a firm's Annual Report for 2022 mentions any information related to margin calls. The first step of the procedure uses a Python code to go through each Annual Report and scrape sentences that contain any variation of at least one of the search keywords in each list of keywords:

search_keywords_1 = [*'margin', 'collateral'*]

search_keywords_2 = [*'call', 'variation', 'initial', 'maintenance', 'require', 'post', 'cost', 'payment', 'expense', 'cover', 'fund', 'outflow', 'spending', 'deposit', 'account', 'broker', 'exchange', 'derivative', 'futures', 'future contract', 'forward', 'options', 'option contract', 'swap', 'cash', 'liquidity'*]

The second step uses Chat GPT-4 to go through the scraped sentences and determine whether a given firm has experienced margin calls during 2022 related to its use of commodity derivatives for hedging. The LLM is given some background information about the task and instructed to look for any details related to margin calls, while ignoring sentences that contain the keywords but discuss other issues (e.g. some mention profit margins or operating margins). Finally, for each company, the procedure outputs a dummy variable that is equal to 1 if Chat GPT-4 determines that the company has experienced a margin call, or 0 otherwise.

A3. Validation of variables composing the Margin Calls Proxy

This table shows the balance sheet variables where firms report cash collateral payments related to margin calls. Out of 36 companies in our sample that mention margin calls in their Annual Report for 2022, 25 explain exactly where the amounts are reported in their balance sheet. We collect this information manually. For each firm, we show the balance sheet line and item where margin amounts are reported, and we include relevant additional clarifications from the text of the Annual Report. In the last column, we mark which of our three chosen variables for the Margin Calls Proxy (Other Current Assets, Other Receivables, and/or Restricted Cash) captures the margin payments of each company.

Company name	Balance sheet line and item	Margin Calls Proxy variable
ALTEO Energy Services Public Limited Company	Current assets and assets held for sale - Other financial assets - Deposits and security deposits - financial assets pledged as collateral for gas forward transactions and electricity trading transactions	Other Current Assets
BKW AG	Current assets - Trade accounts receivable and other receivables - Other financial receivables - collateral deposited with exchanges for the trading of energy derivatives	Other Receivables
BP p.l.c.	Current assets - Cash and cash equivalents - restricted cash balances - amounts required to cover initial margin on trading exchanges and certain cash balances which are subject to exchange controls	Restricted Cash
CEZ, a. s.	Current Assets - Trade Receivables, Net - Margin calls, Collaterals	Other Receivables
Drax Group plc	Current assets - Trade and other receivables and contract assets - Other receivables - Cash collateral payments	Other Receivables
EDP Renováveis, S.A.	Other debtors and other assets - Current - Sundry debtors and other operations - cash collateral held at derivatives exchanges	Other Current Assets
EDP, S.A.	Other debtors and other assets - Current and Non-Current - Guarantees rendered to third parties - collateral requested for transactions in electricity and gas markets	Other Current Assets
Elia Group SA/NV	Current assets - Trade and other receivables - Other receivables - margin calls	Other Receivables

(*continued on next page*)

(*continued*)

Company name	Balance sheet line and item	Margin Calls Proxy variable
EnBW Energie Baden-Württemberg AG	Other assets, Current and Non-current - Miscellaneous assets - contain collateral for exchange-based and over-the-counter trading as well as variation margins	Other Current Assets
Endesa, S.A.	Other financial assets, current and non-current - Loans and other receivables - Financial guarantees recognised as assets - related to trading in organised derivatives markets	Other Receivables
Enel SpA	Current assets - Other current financial assets - Financial assets and cash collateral - cash collateral paid to counterparties for derivatives transactions	Other Current Assets
Engie SA	Current assets - Trade and other receivables, net - Margin calls	Other Receivables
Equinor ASA	Current assets - [Trade and other receivables - Collateral receivables] & [Cash and cash equivalents - Restricted cash, including collateral deposits]	Other Receivables + Restricted Cash
Fortum Oyj	Current assets - Margin receivables	Other Receivables
Iren SpA	Current assets - Sundry assets and other current assets - Other current assets	Other Current Assets
MOL Group	Current assets - Trade and other receivables - other receivables such as margining receivables	Other Receivables
MVV Energie AG	Current assets - Other receivables and assets - Receivables from security deposits for energy trading transactions	Other Receivables
National Grid plc	Current assets - Financial and other investments - include receivables of collateral deposited in relation to derivatives	Other Receivables
Orlen S.A.	Current assets - [Other assets - include margin deposits] & [Cash - Restricted cash - includes funds in margin accounts securing transactions on commodity exchanges]	Other Current Assets + Restricted Cash
PGE Polska Grupa Energetyczna S. A.	Current assets - [Trade and other financial receivables - Deposits, security and collateral - security and transaction deposits and the commodity trading-related guarantees] & [Cash and cash equivalents - includes restricted cash which constitutes collateral for settlements with clearinghouses]	Other Receivables + Restricted Cash
Public Power Corporation S.A.	Current Assets - Other receivables - Electricity and CO2 insurance margin	Other Receivables
RWE Aktiengesellschaft	Current assets - Financial receivables - Collaterals for trading activities	Other Receivables
Shell plc	Current and Non-current Assets - Trade and other receivables - Cash collateral pledged	Other Receivables
SSE plc	Current assets - Trade and other receivables - Cash held as collateral	Other Receivables
TotalEnergies SE	Current Financial Assets - Current deposits beyond three months - include initial margins held as part of activities on organized markets	Other Current Assets

A4. Example of Margin Calls Proxy calculation

Below we show how we calculate our Margin Calls Proxy using Drax Group plc as an example. Drax is a British energy company that engages in the generation and sale of electricity. Their Annual Report for 2022 discloses "cash collateral outflows of £407 million during the year, associated with forward sales of power", which corresponds to approximately €459 million.[20]

Drax reports cash collateral payments in Other Receivables on the balance sheet. Indeed, we find that their Other Receivables as reported by S&P Capital IQ rise sharply by a factor of 128 from 2021-Q2 (before the energy crisis) to 2022-Q3 (when they reach their maximum amount). The other two variables that we use in the calculation of the Margin Calls Proxy do not show similar increases: Restricted Cash is zero, and Other Current Assets (excluding irrelevant components) decrease over the same period.

$$\textit{Margin Calls Proxy}_{Drax} = \max\{\textit{Margins Indicator}_{Drax,t} \mid t \in \textit{[2021-Q3, 2022-Q3]}\} - \textit{Margins Indicator}_{Drax,2021\text{-}Q2} = \text{Margins } \textit{Indicator}_{Drax,2022\text{-}Q3} - \textit{Margins Indicator}_{Drax,2021\text{-}Q2} = [(\text{Other } \textit{Receivables}_{Drax,2022\text{-}Q3} + \text{Restricted } \textit{Cash}_{Drax,2022\text{-}Q3} + \text{Other } \textit{Current Assets}_{Drax,2022\text{-}Q3} - \text{Accrued } \textit{Interest Receivable}_{Drax,2022\text{-}Q3} - \text{Current } \textit{Assets Of Discontinued Operations}_{Drax,2022\text{-}Q3} - \text{Current } \textit{Derivative Assets}_{Drax,2022\text{-}Q3}) - (\text{Other } \textit{Receivables}_{Drax,2021\text{-}Q2} + \text{Restricted } \textit{Cash}_{Drax,2021\text{-}Q2} + \textit{Other Current Assets}_{Drax,2021\text{-}Q2} - \textit{Accrued Interest Receivable}_{Drax,2021\text{-}Q2} - \textit{Current Assets Of Discontinued Operations}_{Drax,2021\text{-}Q2} - \textit{Current Derivative Assets}_{Drax,2021\text{-}Q2})] \; / \; \textit{Total Assets}_{Drax,2021\text{-}Q2} = \text{€M } \textit{678} \; / \; \text{€M } \textit{5832} = \textit{0.116}$$

Drax's Margin Calls Proxy of 0.116 is close to the sample mean (0.11). It corresponds to an estimated €678 million of margins deposited during the energy crisis. Our estimation is higher than the €459 million disclosed in their Annual Report. One possible reason is the different time period on which their calculation is based. The Annual Report compares collateral deposits at the end of 2022 with those at the end of 2021, when margin requirements would already have started rising. In contrast, we estimate the increase in margin deposits from before the start of the crisis to the peak of the liquidity strain, aiming to capture the total margin calls faced by the firm. Another reason for the difference is potential measurement error. Due to limitations that we discuss in the paper, our proxy is imperfect. Although it is an overall useful indicator of liquidity stress, we are not able to measure with precision the euro amounts of margin calls.

A5. Collateral requirements of ETD vs. OTC derivatives

There are three primary ways of entering into a derivatives contract: exchange-traded derivatives (ETD), centrally-cleared over-the-counter (OTC) derivatives, and OTC derivatives which are bilaterally negotiated and traded without involving a central clearing counterparty (CCP). Here we provide a detailed discussion of the different collateral requirements that each of these types of contracts carry.

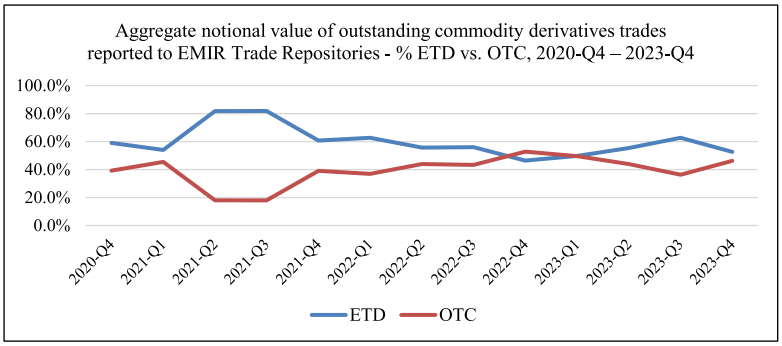
Historically, ETD contracts have been associated with stricter standards of collateralization through daily settlement and mandatory collection of margins by the exchange clearing house. Since the Global Financial Crisis of 2008, G20 leaders have initiated important reforms in derivatives markets meant to reduce systemic risks by tightening rules in OTC trading as well. In 2012, the European Market Infrastructure Regulation (EMIR)[21] introduced rules which, among other things, require all standardized OTC derivatives contracts to be centrally cleared through CCPs. These contracts may be bilaterally negotiated but are executed through a central clearing house similarly to ETDs. As a result, they carry standard initial and variation

[20] Converted at the 31/12/2022 EUR/GBP exchange rate taken from S&P Capital IQ.

[21] Links to the relevant regulation texts can be found here: https://finance.ec.europa.eu/capital-markets-union-and-financial-markets/financial-markets/post-trade-services/derivatives-emir_en

margin requirements. Under the amended EMIR Refit of 2019, the clearing threshold for non-financial counterparties in commodity derivative contracts is EUR 4 billion in gross notional value – above this amount, the entities face an obligation to clear derivatives transactions with a CCP.

Aggregate statistics by EMIR-registered trade repositories of the total notional value of open positions in commodity derivatives suggest that the proportion of OTC over ETD increased somewhat during the energy crisis, reaching 52.8 % in 2022-Q4, and went back down to around 40 % in 2023, as portrayed in the figure below.[22] While there appears to be a partial move towards OTC markets, it is rather limited, temporary, and only happens after the largest margin calls of 2022-Q3. This is consistent with the evidence in Chart A.4 by the ECB Financial Stability Review which shows gross exposures in energy derivatives specifically. From the beginning to the end of 2022, we can observe a slight drop (<10 %) in the fraction of all energy derivatives that are ETD in favor of OTC; around 60 % of contracts remain on-exchange.



Nevertheless, the proportion of energy derivatives contracts that are OTC reaches around 40 %, which is not insignificant. Therefore, we need to discuss the collateral requirements for these contracts.

If an OTC contract is not centrally cleared, risk-mitigation techniques must be applied according to Article 11 of EMIR. Counterparties are required to "mark-to-market on a daily basis the value of outstanding contracts" and "have risk-management procedures that require the timely, accurate and appropriately segregated exchange of collateral", with Regulation 2016/2251 further clarifying that this includes the exchange of both initial and variation margins. These requirements apply to all financial counterparties and to any non-financial counterparties (including energy companies) with positions that exceed the clearing threshold. Even when it comes to companies that are below the clearing threshold, it is likely that many still need to make collateral payments. For example, suppose an energy firm hedges through OTC commodity derivatives but its positions do not exceed the EUR 4 billion threshold. If the firm negotiates its derivative contracts with a bank dealer or another financial institution, then that financial institution is under obligation to request collateral payments from the firm. The same holds if the firm's counterparty is another non-financial corporation whose positions do exceed the clearing threshold. The only way for the firm to circumvent the collateral requirements is to enter into a bilateral agreement with another non-financial firm that is also below the threshold, in which case neither of them falls under the EMIR obligation. While this is possible, it would expose the firms to considerably higher counterparty credit risk. Bilateral OTC agreements usually incorporate capital charges to protect counterparties from such risk. This means that even in cases where central clearing is voluntary, there may be incentives to clear transactions through a CCP to reduce the risk as well as the capital requirements, including taking advantage of netting opportunities (see discussions of clearing incentives in a 2014 report by the Bank for International Settlements (BIS) OTC Derivatives Assessment Team[23] and in Bellia et al., 2024).

Despite the potential benefits of central clearing, it remains relatively uncommon for OTC derivatives on commodities. Commodity contracts tend to involve customization and physical settlement more often than other classes of OTC derivatives, resulting in operational challenges for CCPs. Only 9 % of OTC commodity derivatives were centrally cleared over our sample period, according to a report by ESMA.[24]

Yet, there is evidence that even when central clearing and collection of margins is not mandatory, many OTC contracts are still voluntarily collateralized. For example, the International Swaps and Derivatives Association (ISDA) Margin Survey 2010[25] reports that 64 % of non-centrally cleared OTC commodity derivatives trades included collateral agreements, even though at the time collateral regulations had not been introduced yet. In addition, the survey reports that 83 % of all collateral was delivered in cash.

While the above discussion pertains to the EU jurisdiction, equivalent regulations have been introduced globally. The United Kingdom has adopted the so-called UK EMIR,[26] and the European Commission has determined that the regulatory regimes of many other non-EU countries, including the United States, are equivalent to the EMIR framework.[27] In addition, a global policy framework established by the Basel Committee on Banking Supervision (BCBS) and the International Organization of Securities Commissions (IOSCO) sets standards for mandatory exchange of margins in non-centrally cleared OTC derivatives, following a phased-in schedule starting in 2016, for all financial firms and systemically important non-financial entities.[28]

[22] Since 2021, there are four registered trade repositories (TRs) in the EU to which commodity derivatives trades are reported under EMIR. Our calculations are based on publicly available data from three of them (DTCC, LSEG/UnaVista, and KDPW); the fourth one (REGIS) does not provide historical records, but this is unlikely to affect our results as its market share is relatively low (some estimates of the TRs' market shares by the European Systemic Risk Board can be found here: https://www.esrb.europa.eu/pub/pdf/occasional/20160922_occasional_paper_11.en.pdf).

[23] https://www.bis.org/publ/othp21.pdf

[24] https://www.esma.europa.eu/sites/default/files/2023-12/ESMA50-524821-2930_EU_Derivatives_Markets_2023.pdf

[25] https://www.isda.org/a/UeiDE/isda-margin-survey-2010.pdf

[26] https://www.fca.org.uk/markets/uk-emir

[27] https://finance.ec.europa.eu/eu-and-world/equivalence-non-eu-financial-frameworks_en

[28] https://www.bis.org/bcbs/publ/d317.htm

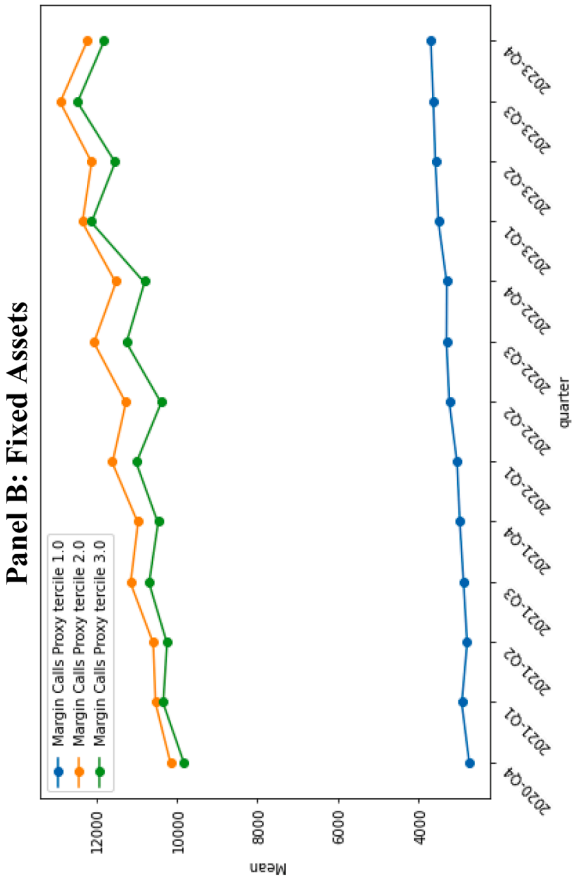
A6. Leverage and hedging

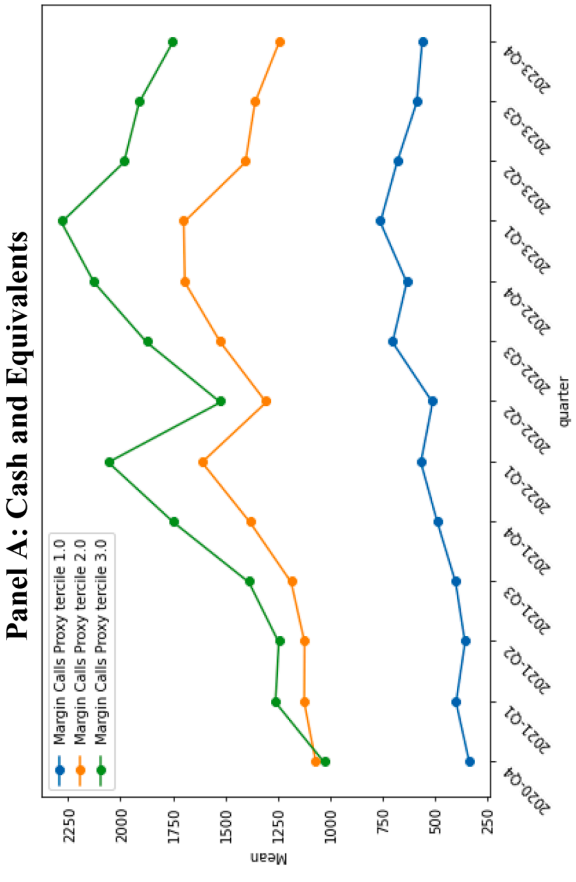
This table reports the differences in means of variables related to hedging intensity between firms with high leverage (above the median) and other firms (below the median) matched by market capitalization, and the corresponding t-statistics. We report results using five different measures of leverage, taken at the end of 2021-Q2 (before the margin calls). The variable (Current) Derivatives Positions refers to the total fair value of (current) derivatives, for those firms that report them at the end of 2021-Q2, scaled by Total Assets. The commodity price betas are obtained by regressing each company's daily stock returns on the commodity returns from 1 January 2015 to 30 June 2021 (refer to Section 2.3 for details on the methodology).

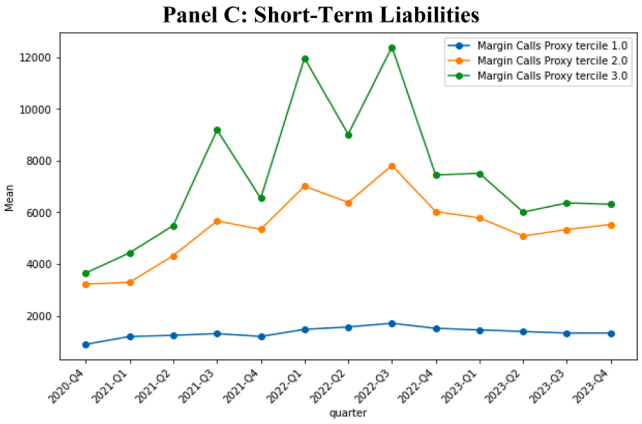
	Difference between firms with high leverage and other firms matched by size									
Leverage measure:	**Total Debt**		**Long-Term Debt**		**Bank Debt**		**Term Loans**		**Undrawn Revolving Credit**	
Variable	**mean diff.**	**t-stat**	**mean diff.**	**t-stat**	**mean diff.**	**t-stat**	**mean diff.**	**t-stat**	**mean diff.**	**t-stat**
Derivatives Positions / Total Assets	−3.5 %	*−1.13*	−2.3 %	*−0.98*	0.2 %	*0.05*	0.6 %	*0.16*	−1.2 %	*−0.42*
Current Derivatives Positions / Total Assets	−3.6 %	*−1.31*	−2.3 %	*−1.03*	−0.4 %	*−0.17*	−0.3 %	*−0.11*	−1.3 %	*−0.51*
Oil Price beta	0.02	*0.72*	0.02	*0.63*	−0.01	*−0.22*	0.00	*0.05*	0.03	*1.02*
Gas Price beta	0.00	*0.47*	0.00	*−0.23*	−0.01	*−1.22*	0.00	*−0.90*	0.01	*0.81*
Power Price beta	0.00	*0.11*	0.00	*−0.43*	0.00	*−1.41*	0.00	*−1.20*	0.00	*0.41*

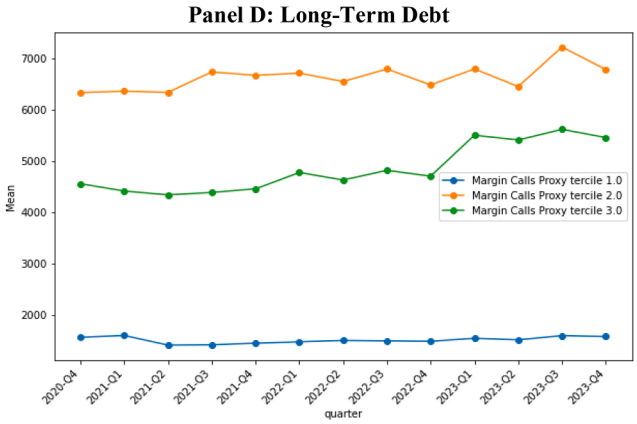
A7. Time-series variation in firm financials by size of margin calls experienced

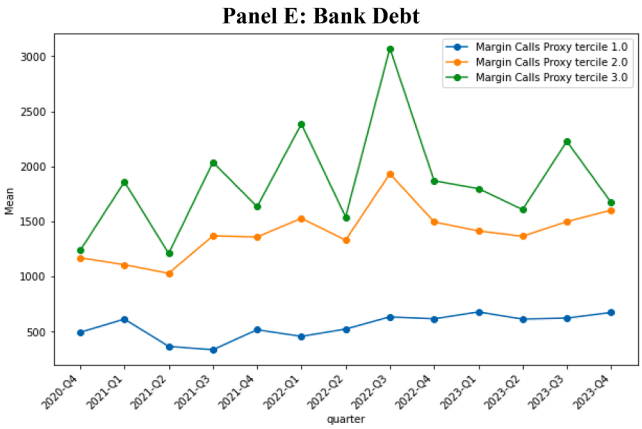
The following figures show the variation in key financial statement items and financial ratios of the companies in our sample from 2020-Q4 to 2023-Q4. We split companies into three terciles based on the proxied size of margin calls experienced (refer to Section 2.3 for details about our Margin Calls Proxy). The blue line plots average values across firms in the bottom tercile that experienced the smallest margin calls; the orange line is for the middle tercile; and the green line plots averages in the top tercile of firms that experienced the largest margin calls. Fixed Assets refer to Net Property, Plant, and Equipment (PP&E). Short-Term Liabilities are Current Liabilities excluding Derivatives Liabilities. Cost of Debt is calculated as a weighted average of the yields on bonds issued by a company. Profit Margin (before tax) is equal to Earnings Before Tax (EBT)/Total Revenue. Variables are winsorized each quarter at the 5th and 95th percentiles.

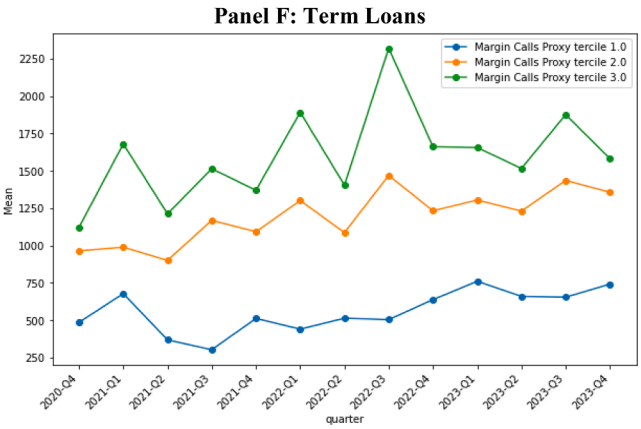

Panel B: Fixed Assets
Margin Calls Proxy tercile 1.0
Margin Calls Proxy tercile 2.0
Margin Calls Proxy tercile 3.0
Mean
quarter
12000
10000
8000
6000
4000
2020-Q4
2021-Q1
2021-Q2
2021-Q3
2021-Q4
2022-Q1
2022-Q2
2022-Q3
2022-Q4
2023-Q1
2023-Q2
2023-Q3
2023-Q4

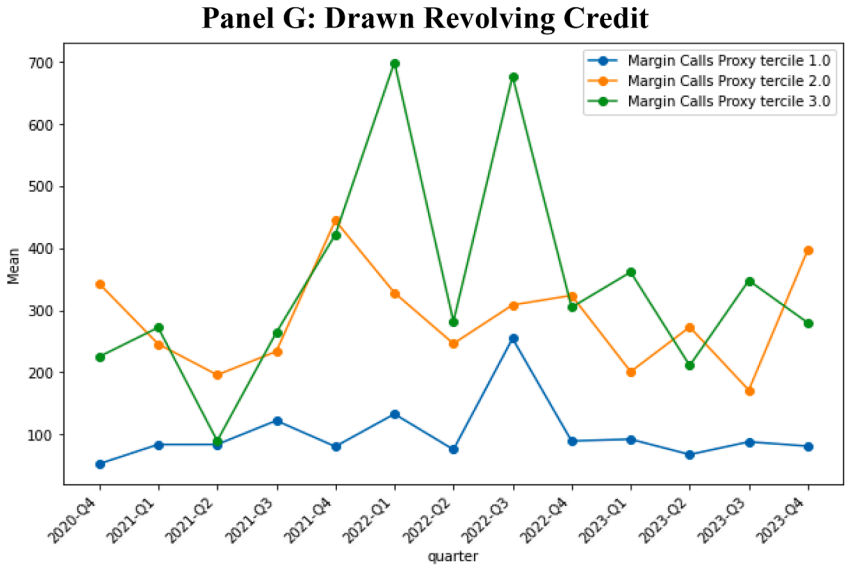

Panel A: Cash and Equivalents
Margin Calls Proxy tercile 1.0
Margin Calls Proxy tercile 2.0
Margin Calls Proxy tercile 3.0
Mean
quarter
2250
2000
1750
1500
1250
1000
750
500
250
2020-Q4
2021-Q1
2021-Q2
2021-Q3
2021-Q4
2022-Q1
2022-Q2
2022-Q3
2022-Q4
2023-Q1
2023-Q2
2023-Q3
2023-Q4

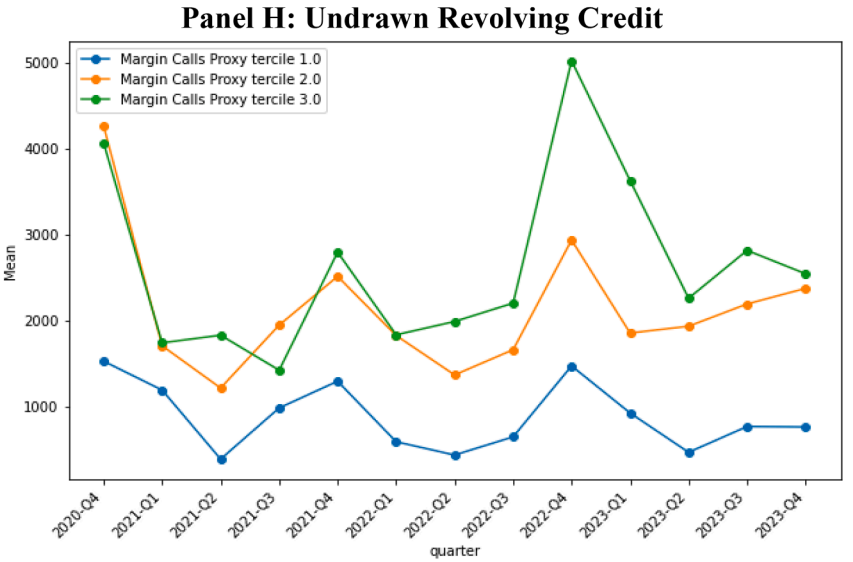

Panel C: Short-Term Liabilities
Margin Calls Proxy tercile 1.0
Margin Calls Proxy tercile 2.0
Margin Calls Proxy tercile 3.0
Mean
quarter

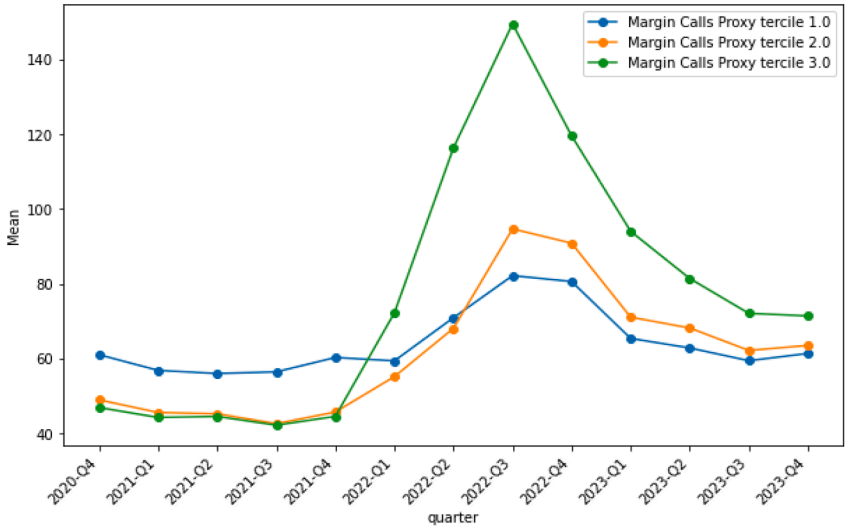

Panel D: Long-Term Debt
Margin Calls Proxy tercile 1.0
Margin Calls Proxy tercile 2.0
Margin Calls Proxy tercile 3.0
Mean
quarter

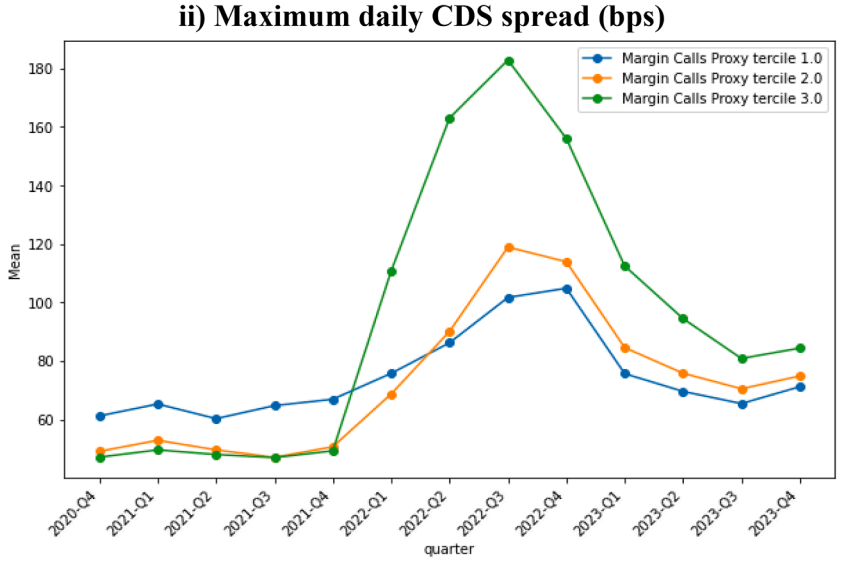

Panel E: Bank Debt
Margin Calls Proxy tercile 1.0
Margin Calls Proxy tercile 2.0
Margin Calls Proxy tercile 3.0
Mean
quarter

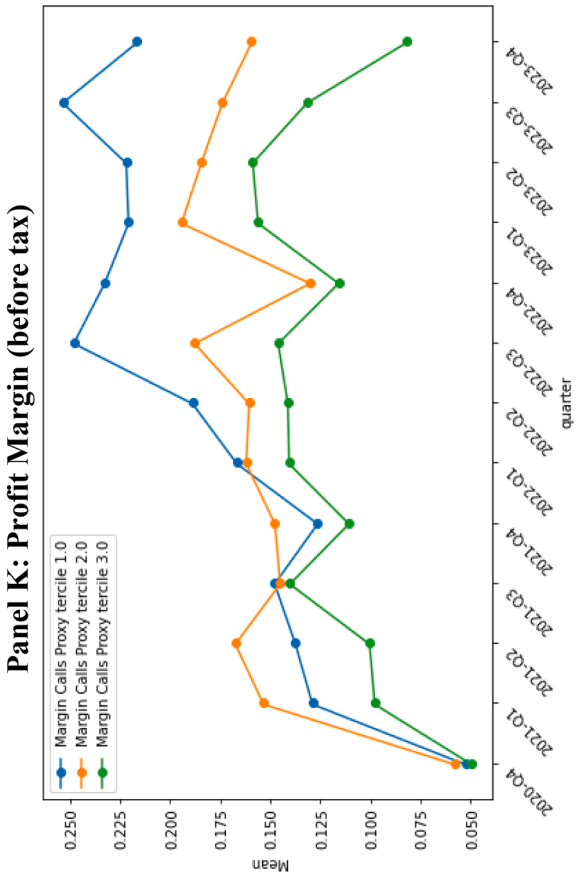

Panel F: Term Loans
Margin Calls Proxy tercile 1.0
Margin Calls Proxy tercile 2.0
Margin Calls Proxy tercile 3.0
Mean
quarter

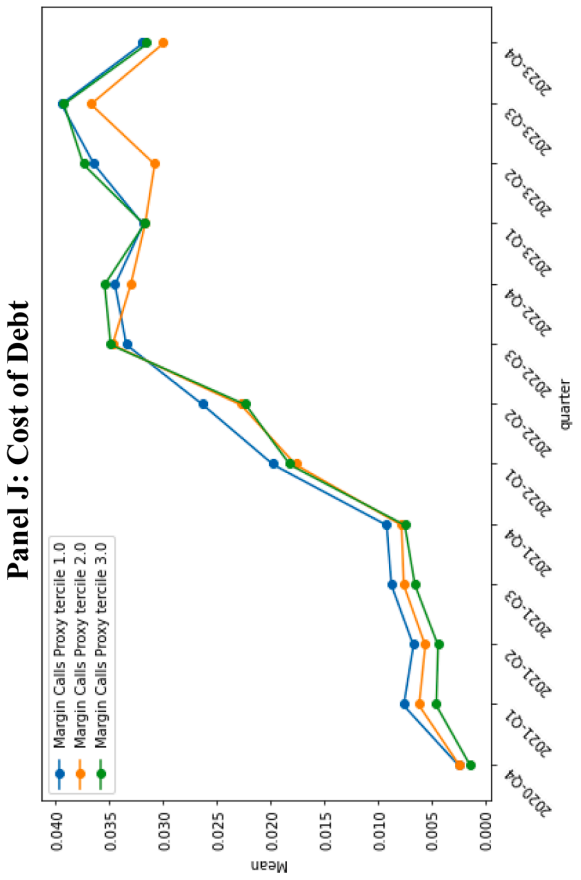

Panel G: Drawn Revolving Credit
Margin Calls Proxy tercile 1.0
Margin Calls Proxy tercile 2.0
Margin Calls Proxy tercile 3.0
Mean
quarter
2020-Q4
2021-Q1
2021-Q2
2021-Q3
2021-Q4
2022-Q1
2022-Q2
2022-Q3
2022-Q4
2023-Q1
2023-Q2
2023-Q3
2023-Q4


Panel H: Undrawn Revolving Credit
Margin Calls Proxy tercile 1.0
Margin Calls Proxy tercile 2.0
Margin Calls Proxy tercile 3.0
Mean
quarter


Panel I: CDS Spreads: i) Average daily CDS spread (bps)
Margin Calls Proxy tercile 1.0
Margin Calls Proxy tercile 2.0
Margin Calls Proxy tercile 3.0
Mean
quarter


ii) Maximum daily CDS spread (bps)
Margin Calls Proxy tercile 1.0
Margin Calls Proxy tercile 2.0
Margin Calls Proxy tercile 3.0
Mean
quarter


Panel K: Profit Margin (before tax)
Mean
quarter
Margin Calls Proxy tercile 1.0
Margin Calls Proxy tercile 2.0
Margin Calls Proxy tercile 3.0
0.250
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.050
2020-Q4
2021-Q1
2021-Q2
2021-Q3
2021-Q4
2022-Q1
2022-Q2
2022-Q3
2022-Q4
2023-Q1
2023-Q2
2023-Q3
2023-Q4


Panel J: Cost of Debt
Mean
quarter
Margin Calls Proxy tercile 1.0
Margin Calls Proxy tercile 2.0
Margin Calls Proxy tercile 3.0
0.040
0.035
0.030
0.025
0.020
0.015
0.010
0.005
0.000
2020-Q4
2021-Q1
2021-Q2
2021-Q3
2021-Q4
2022-Q1
2022-Q2
2022-Q3
2022-Q4
2023-Q1
2023-Q2
2023-Q3
2023-Q4

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